US 20250276621A1

(19) United States
(12) Patent Application Publication
(10) Pub. No.: US 2025/0276621 A1
(43) Pub. Date: Sep. 4, 2025

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(54) **CONVEYANCE SEAT**

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(21) Appl. No.: **19/211,670**

(22) Filed: **May 19, 2025**

Related U.S. Application Data

(63) Continuation of application No. 18/648,531, filed on Apr. 29, 2024, now Pat. No. 12,311,817, which is a continuation of application No. 18/024,985, filed on Mar. 7, 2023, filed as application No. PCT/JP2021/033002 on Sep. 8, 2021, now Pat. No. 11,999,275.

(60) Provisional application No. 63/075,944, filed on Sep. 9, 2020, provisional application No. 63/082,066, filed on Sep. 23, 2020.

Publication Classification

(51) **Int. Cl.**
B60N 2/427 (2006.01)
B60N 2/68 (2006.01)
B60R 21/207 (2006.01)

(52) **U.S. Cl.**
CPC ***B60N 2/427*** (2013.01); ***B60N 2/68*** (2013.01); ***B60R 21/207*** (2013.01)

(57) **ABSTRACT**

Provided is a conveyance seat equipped with a side airbag device with which an airbag can be inflation-deployed in a more stable state. The conveyance seat includes a seat back and a side airbag device attached to a side portion of the seat back. The seat back includes a back frame having a side frame and a side support member attached to the front part of the side frame and protruding to the seat front side beyond the side frame. The side airbag device has a first airbag inflated on the outside surface side of the side frame and a second airbag inflated on the inside surface side of the side frame. The second airbag passes through a gap formed between the side frame and the side support member in the seat front to back direction and is inflated on the inside surface side of side frame.

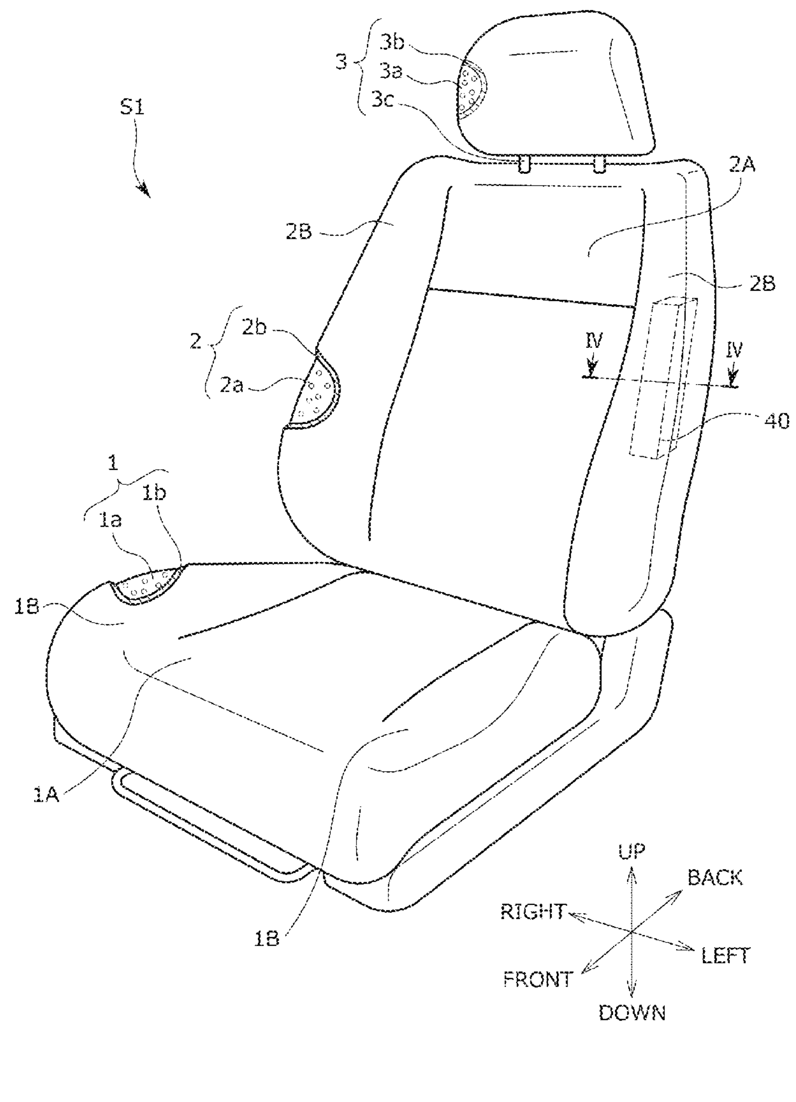
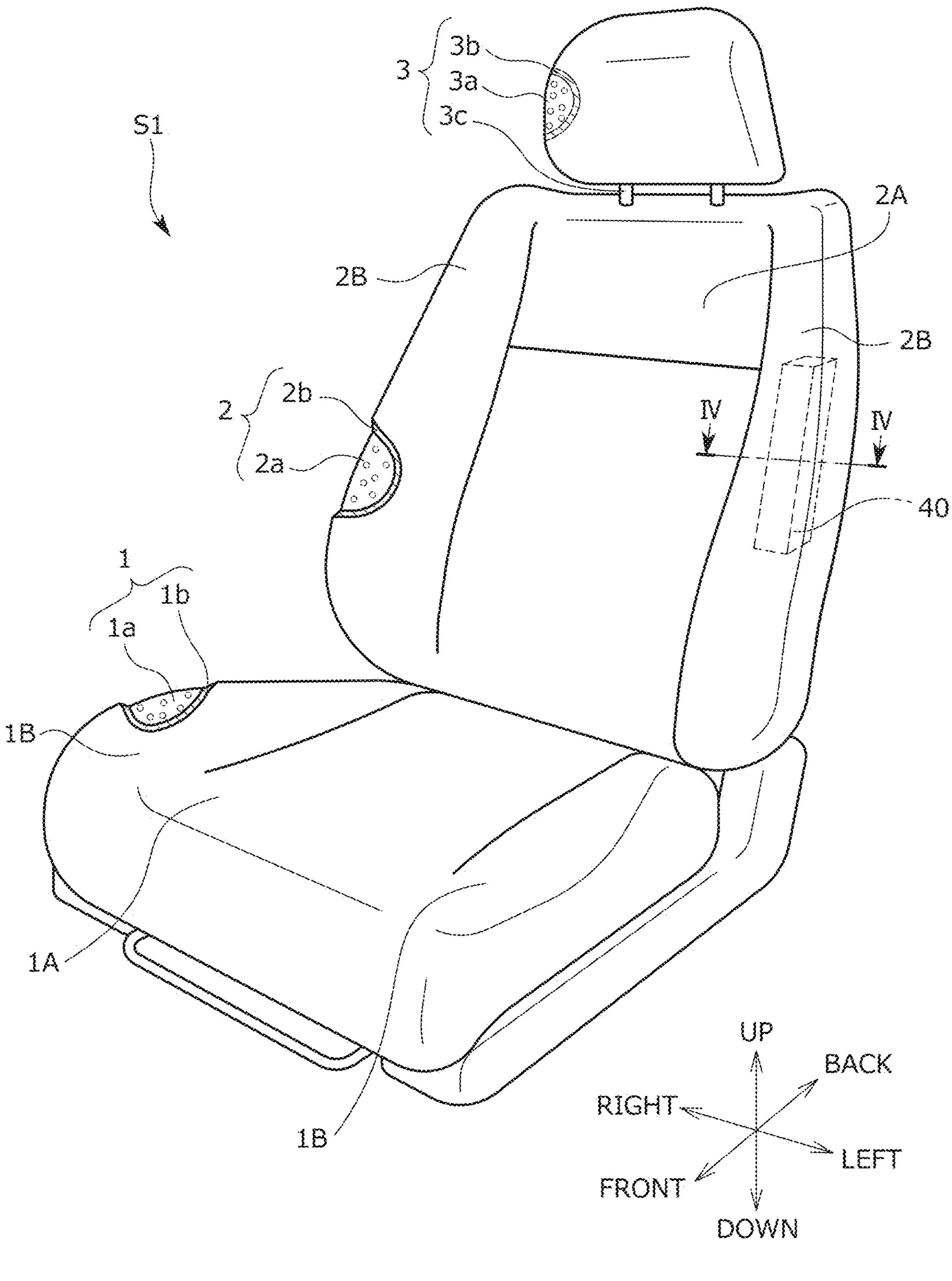


FIG. 1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3b
3a
3c
S1
2A
2B
2B
2
2b
2a
IV
IV
40
1
1b
1a
1B
1A
1B
UP
BACK
RIGHT
LEFT
FRONT
DOWN

FIG. 2

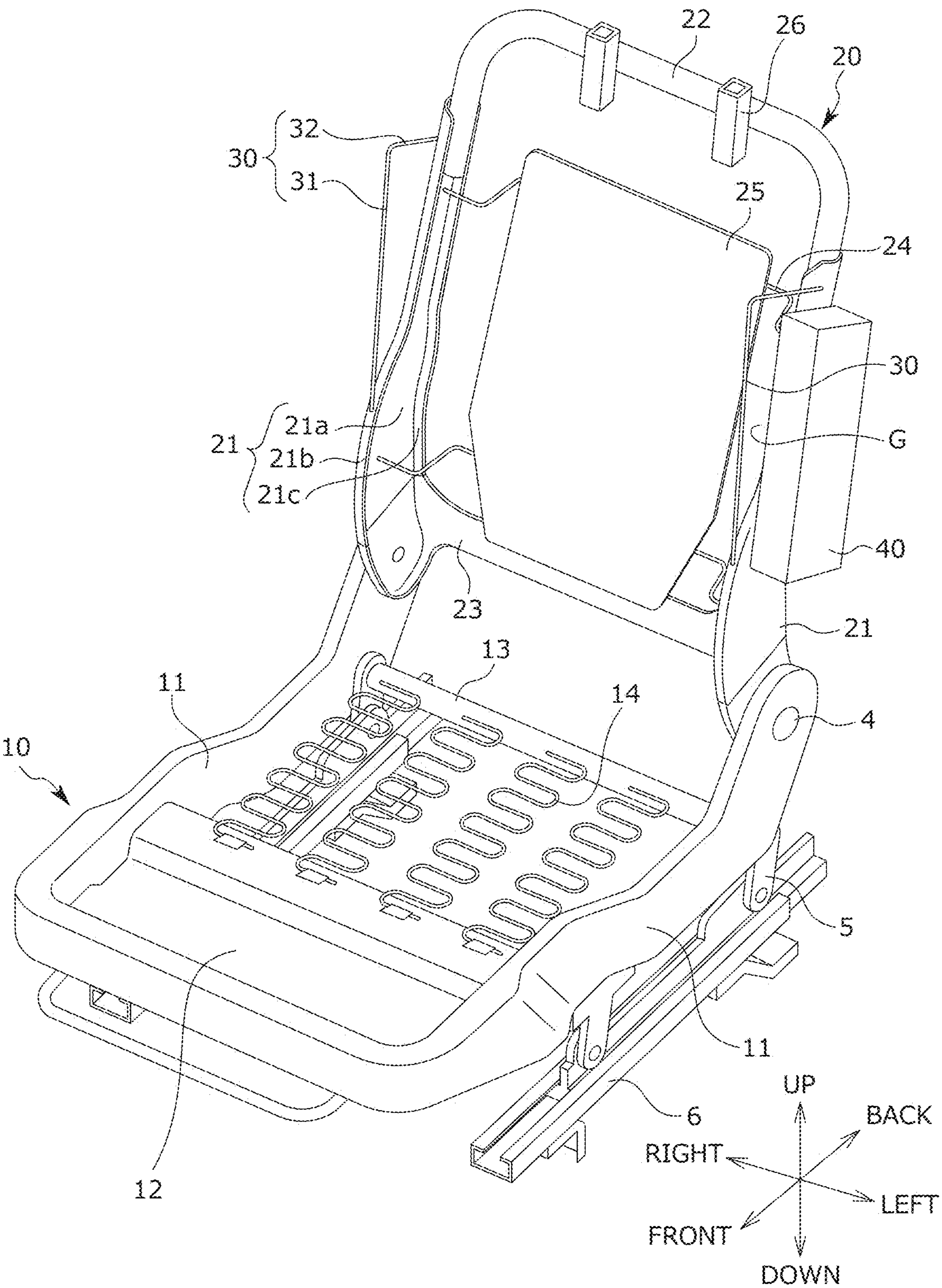

22
26
20
32
30
31
25
24
30
G
21a
21
21b
21c
40
23
21
13
11
14
4
10
5
11
6
12
UP
BACK
RIGHT
LEFT
FRONT
DOWN

FIG. 3A

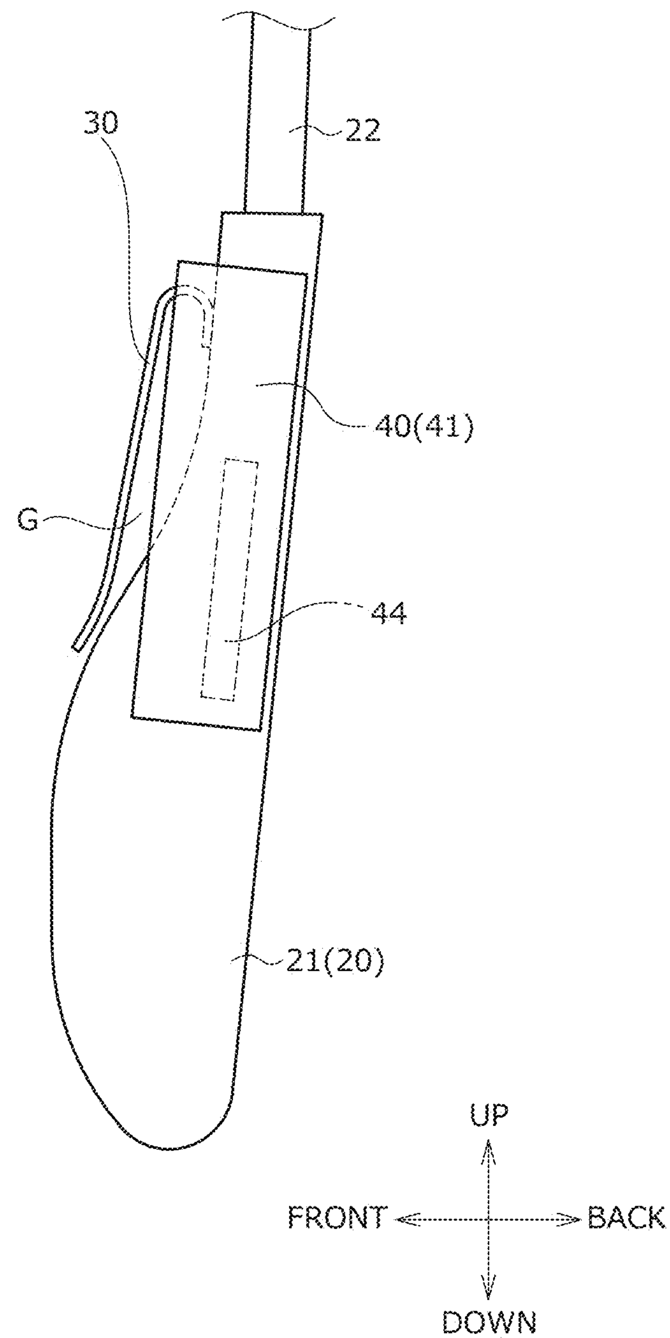


FIG. 3B

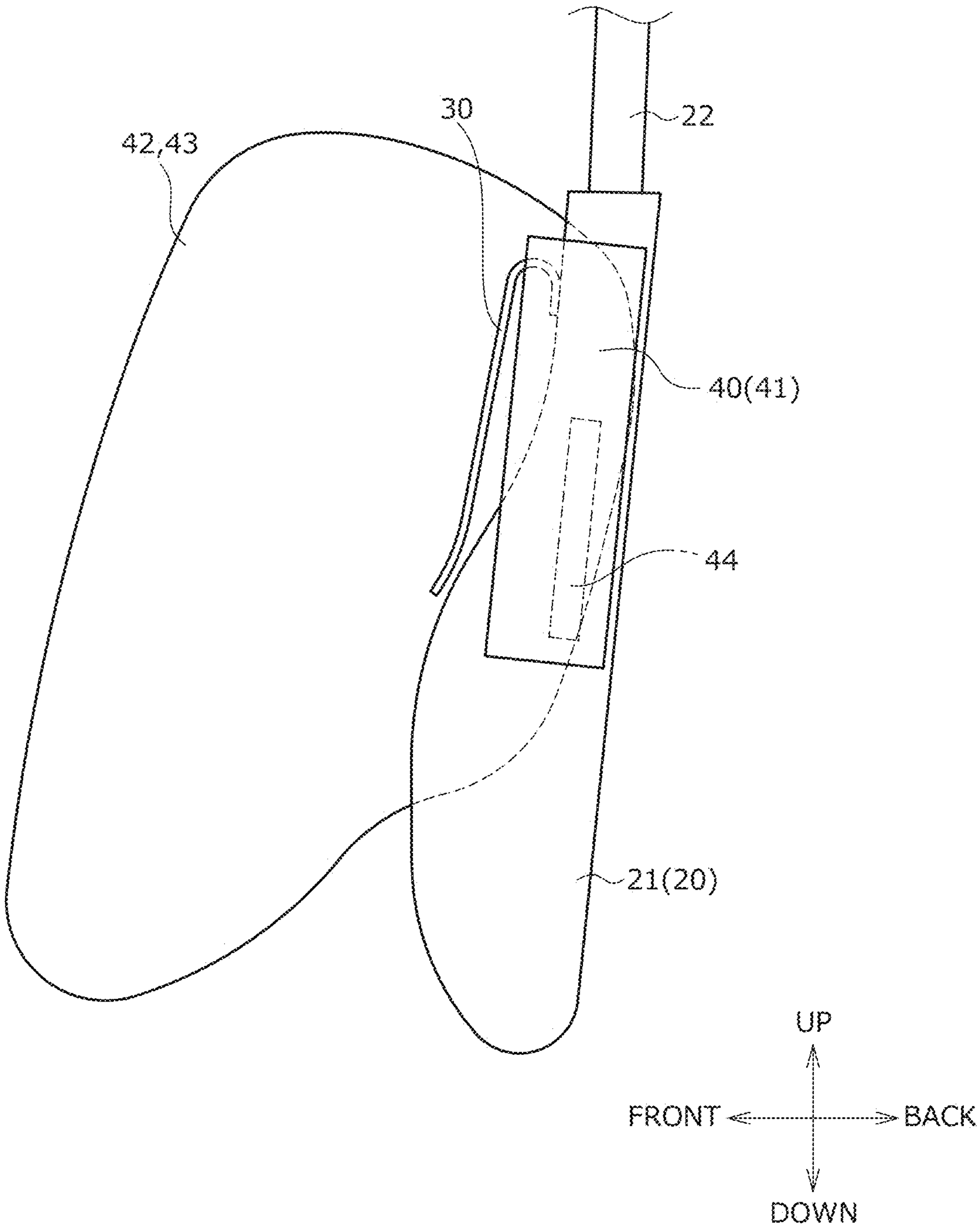


FIG. 4A

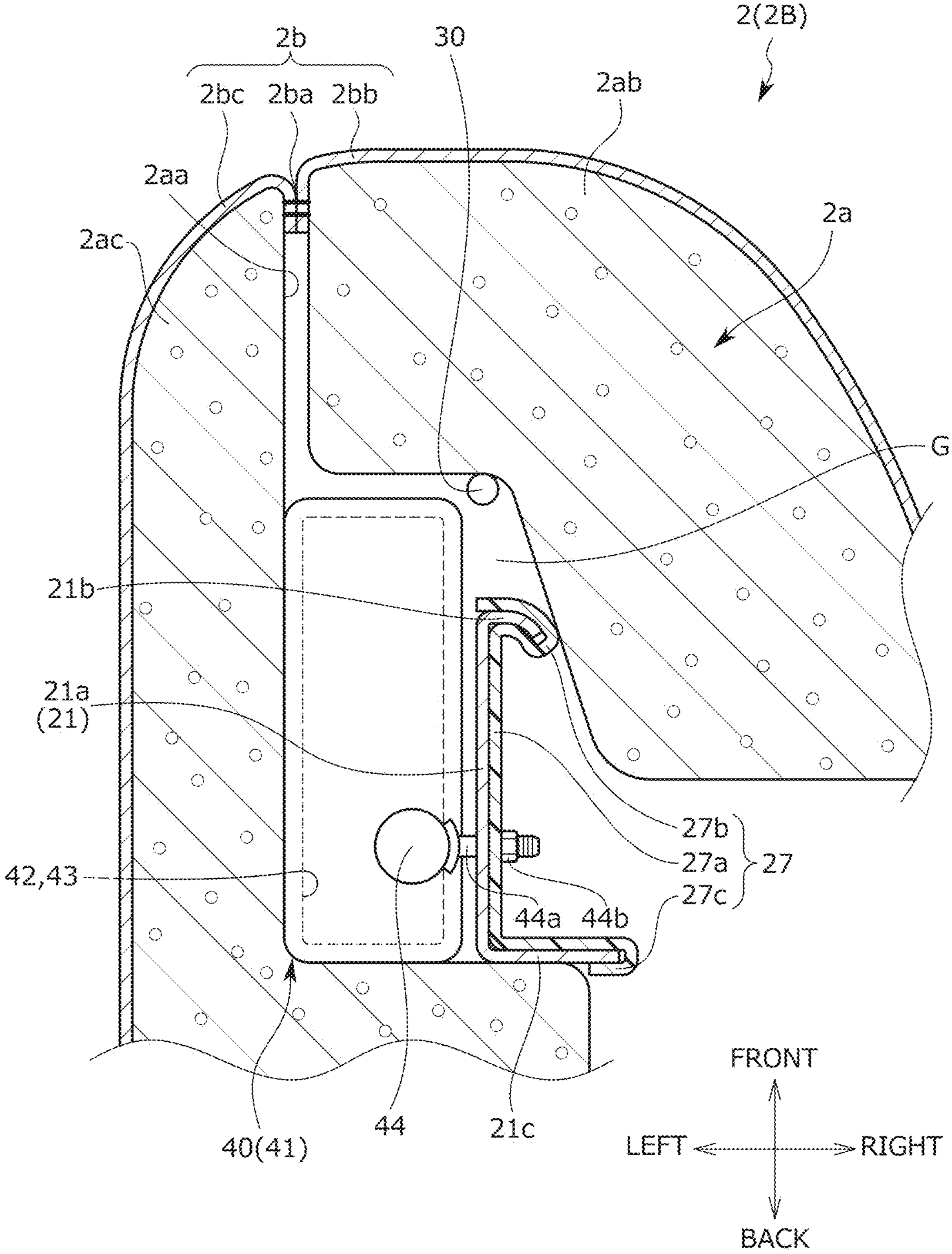

2(2B)
2b
30
2bc
2ba
2bb
2ab
2aa
2ac
2a
G
21b
21a
(21)
27b
27a
27
27c
42,43
44a
44b
40(41)
44
21c
FRONT
LEFT
RIGHT
BACK

FIG. 4B

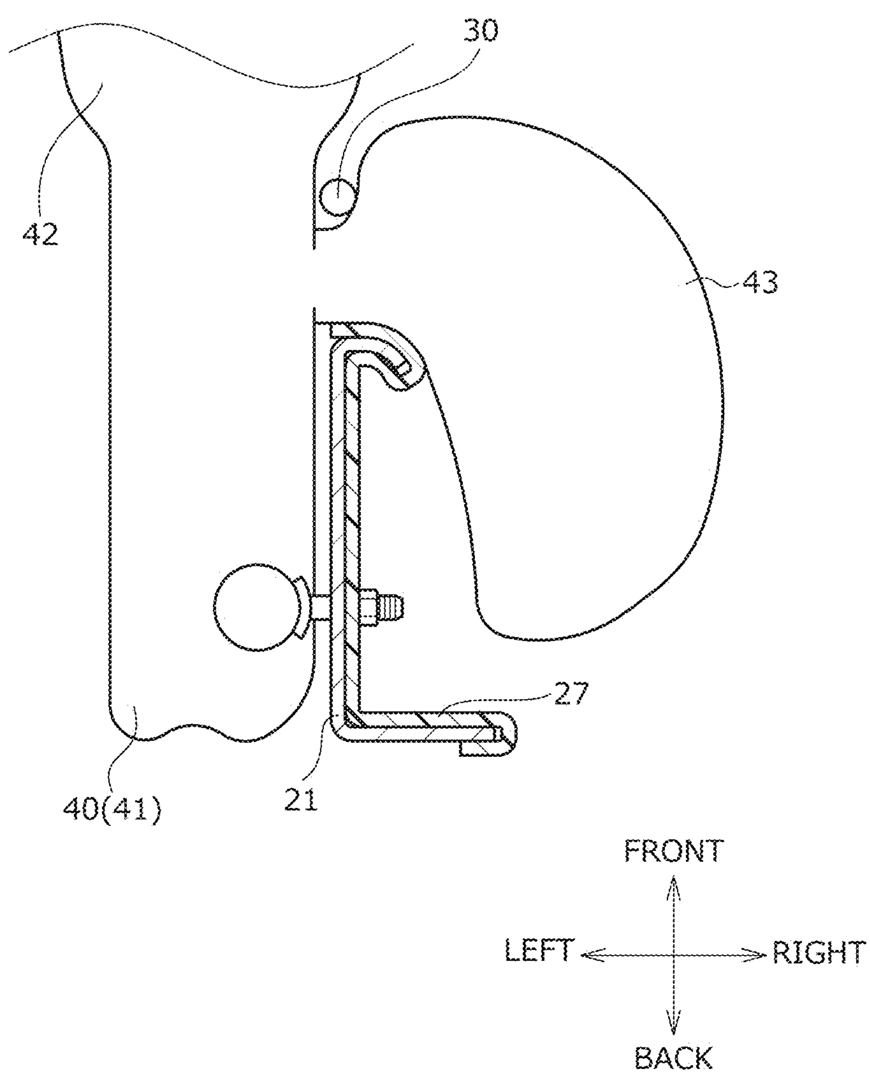


FIG. 5

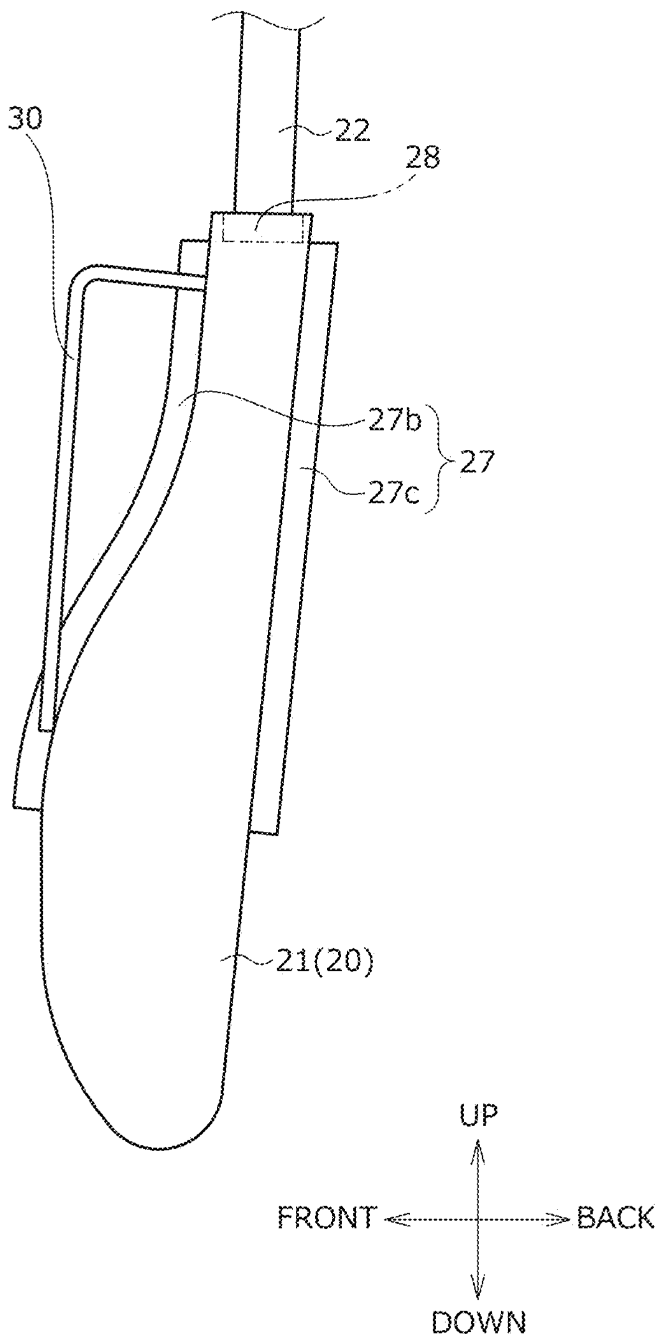


FIG. 6

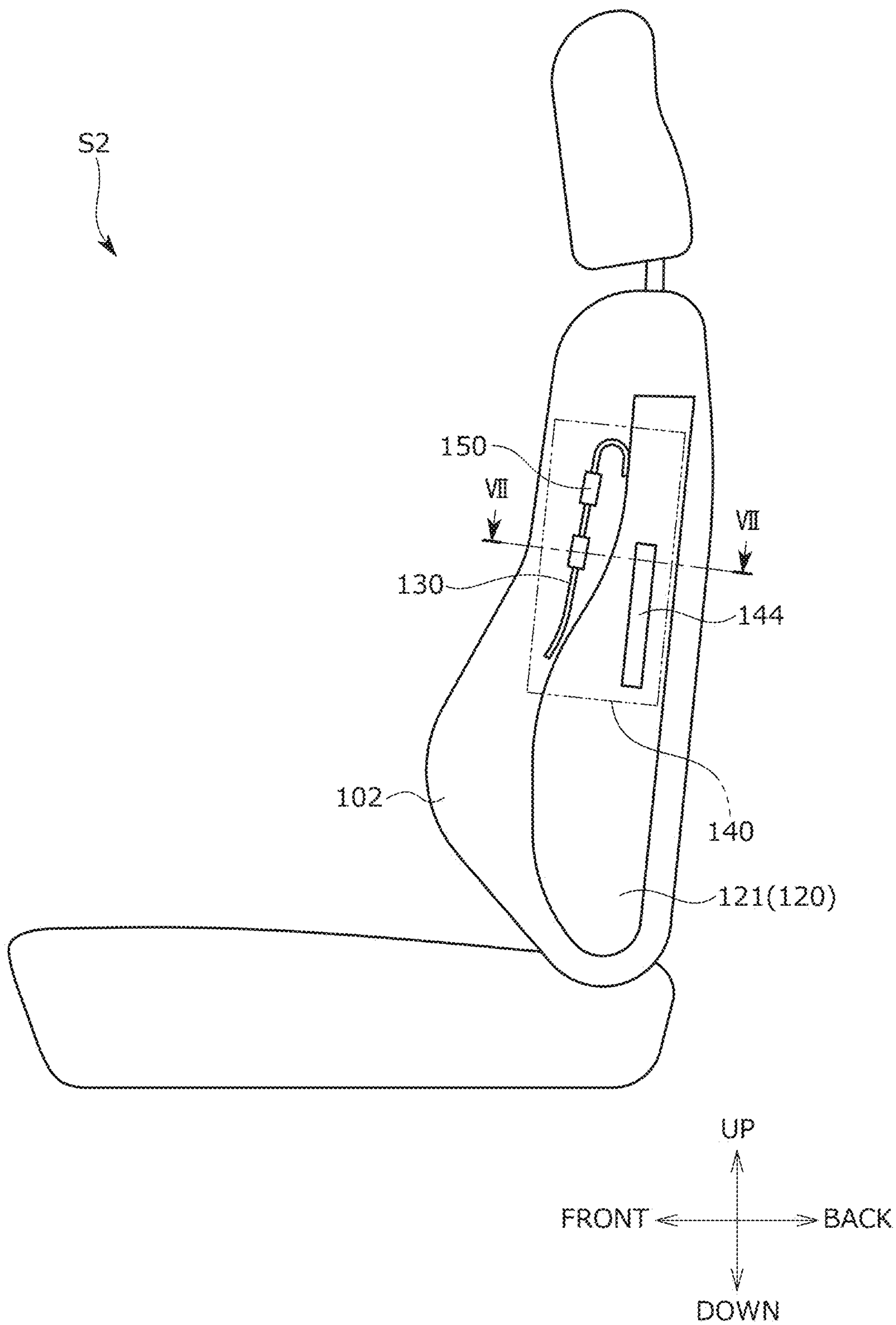
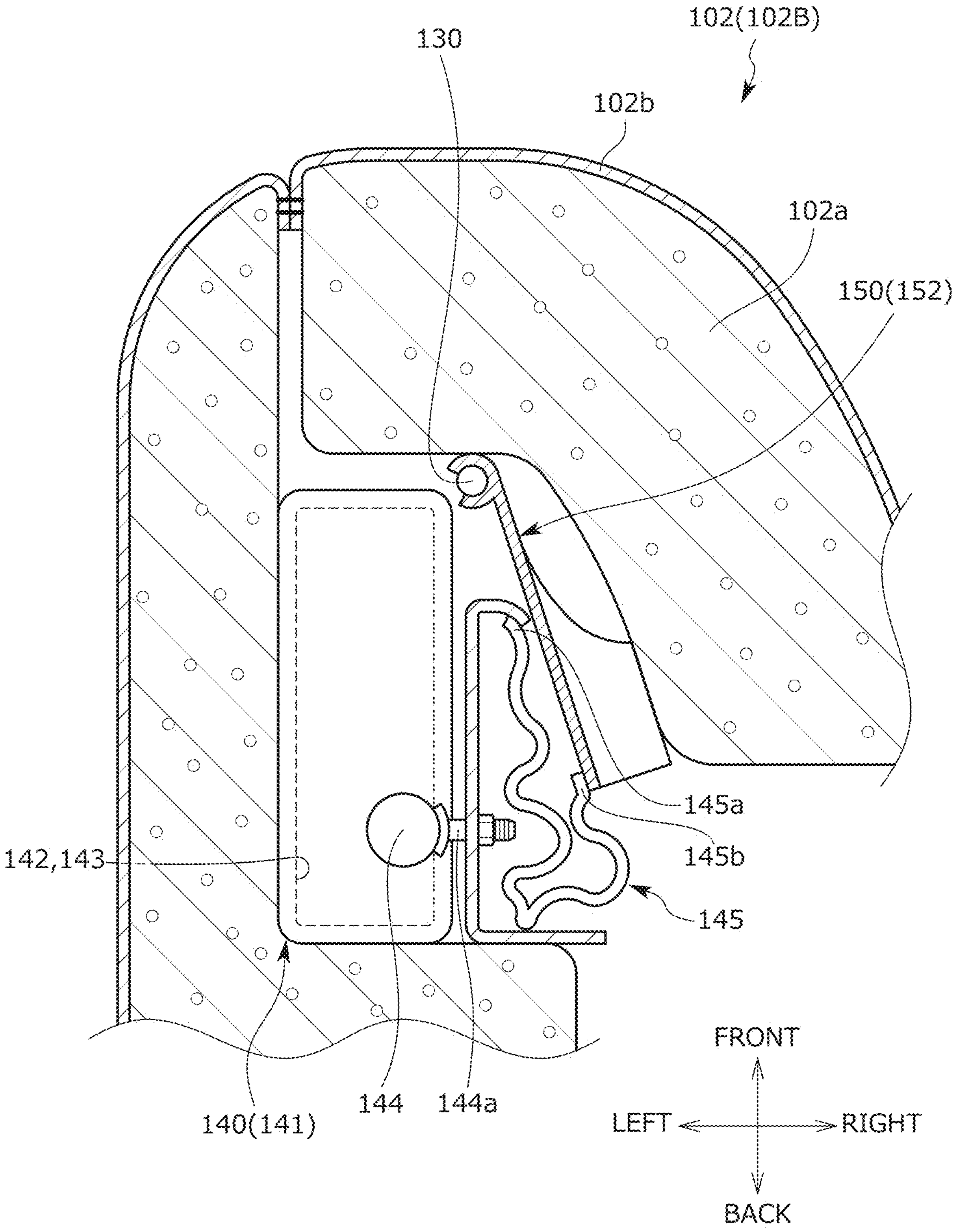


FIG. 7A



102(102B)
130
102b
102a
150(152)
145a
145b
142,143
145
140(141)
144
144a
FRONT
LEFT
RIGHT
BACK

FIG. 7B

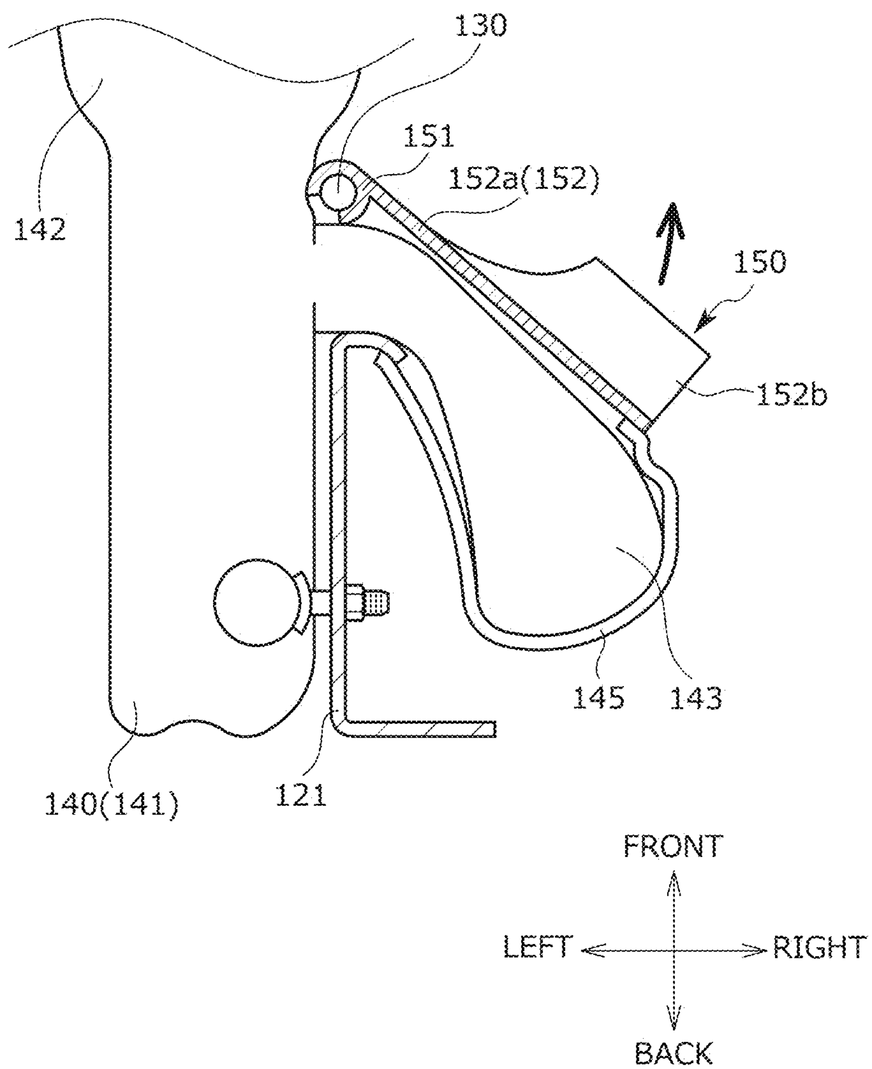

130
151
152a(152)
142
150
152b
145
143
140(141)
121
FRONT
LEFT
RIGHT
BACK

FIG. 8

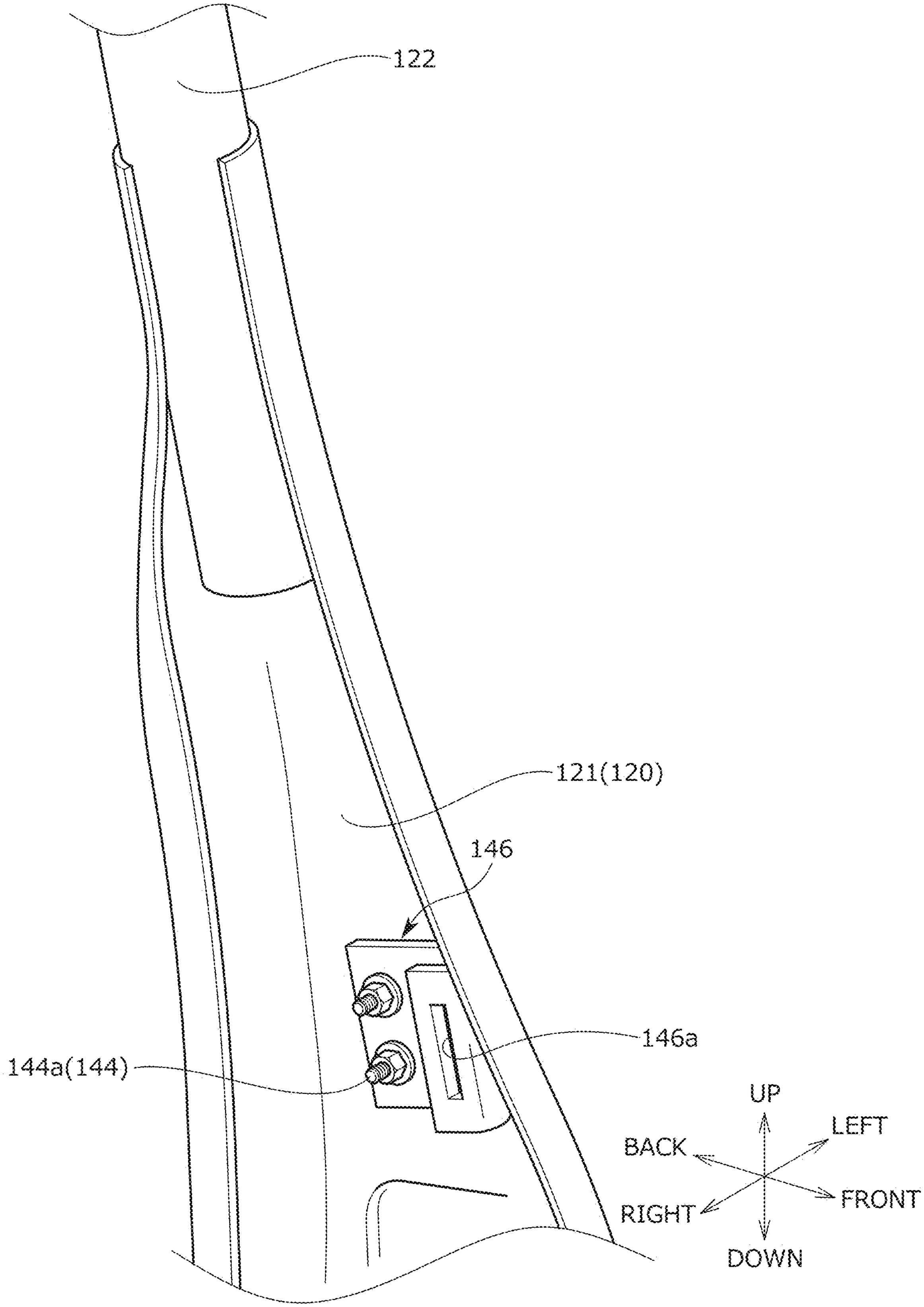


FIG. 9

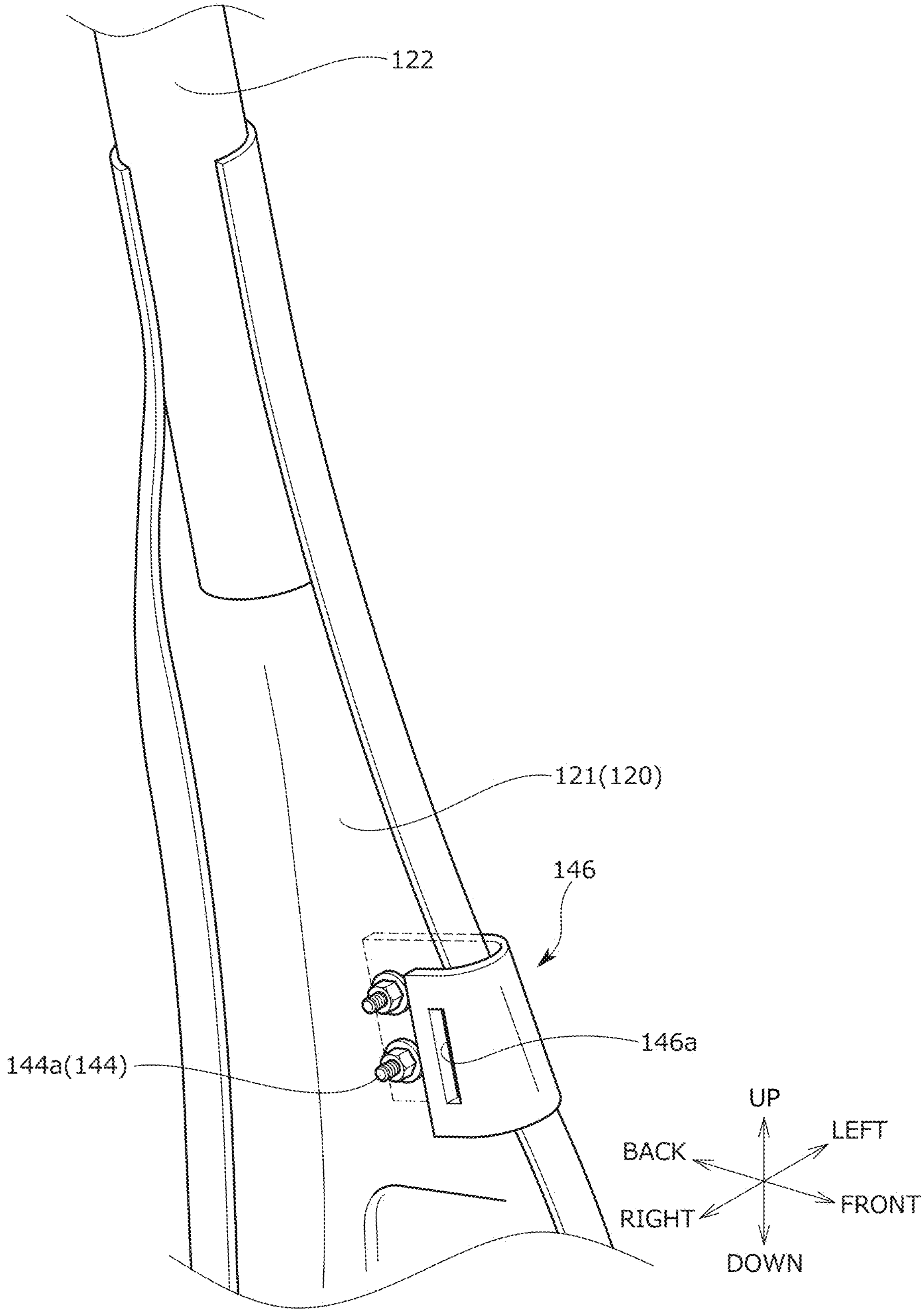


FIG. 10

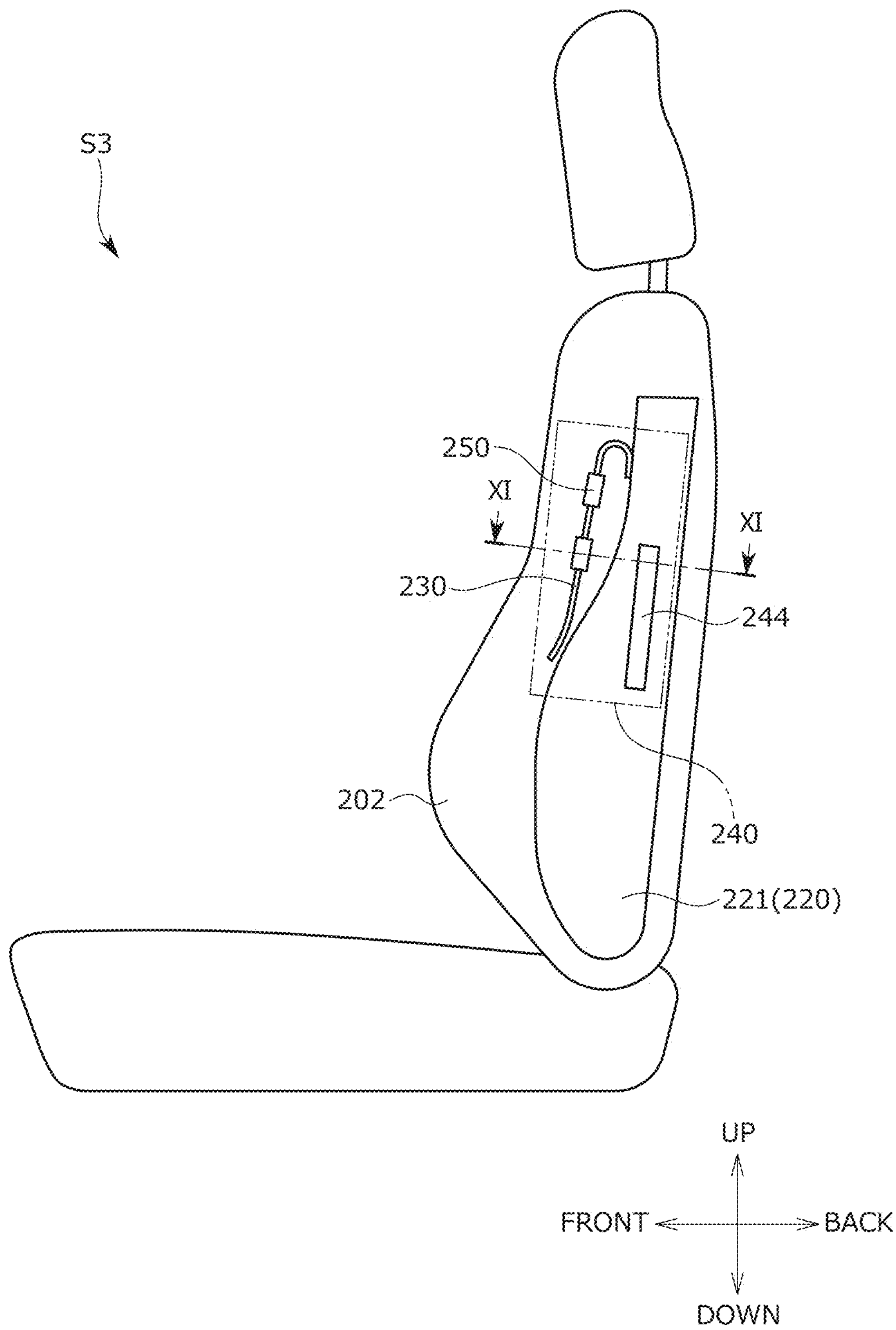


FIG. 11A

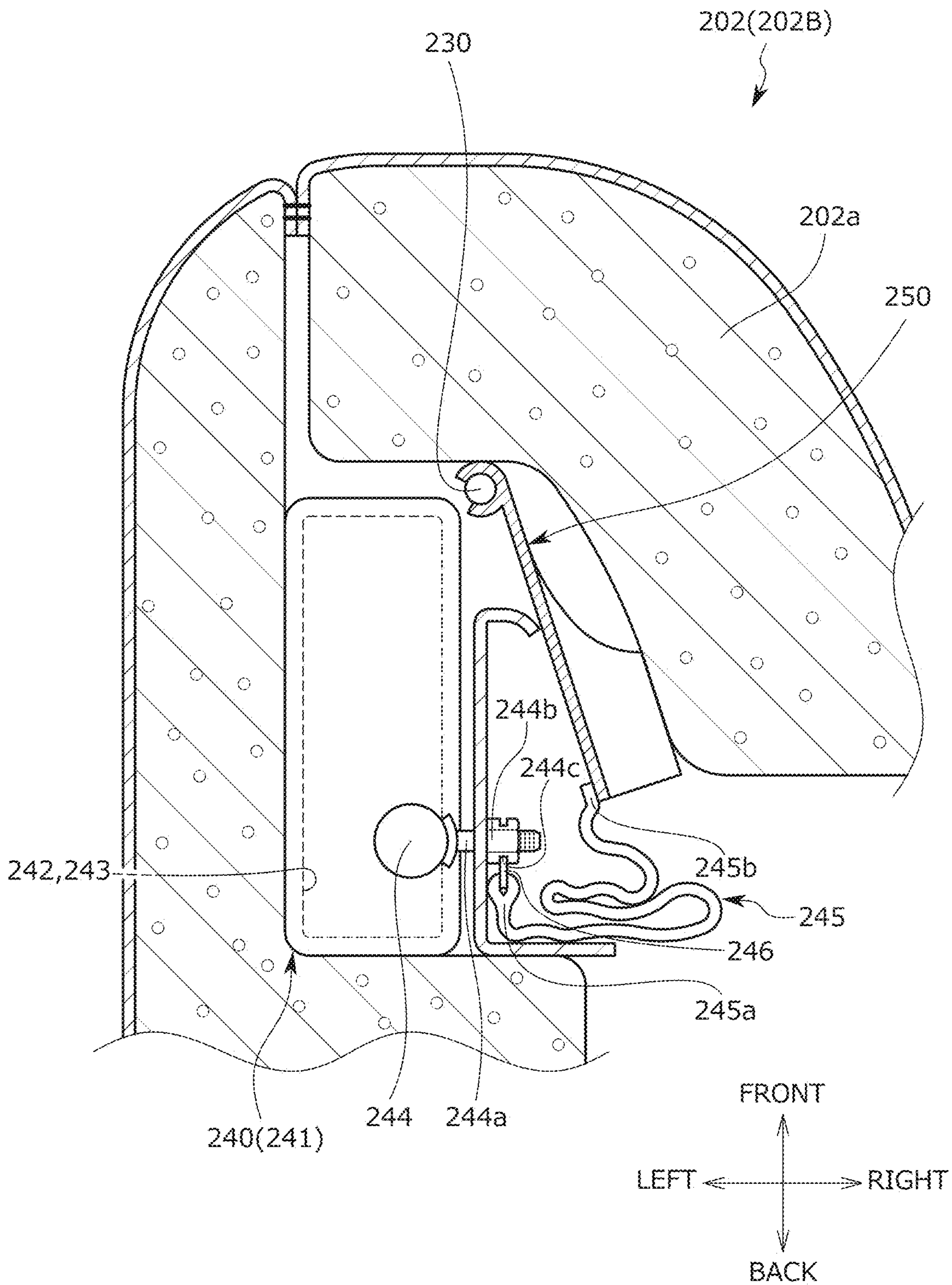

202(202B)
230
202a
250
244b
244c
245b
242,243
245
246
245a
240(241)
244
244a
FRONT
LEFT
RIGHT
BACK

FIG. 11B

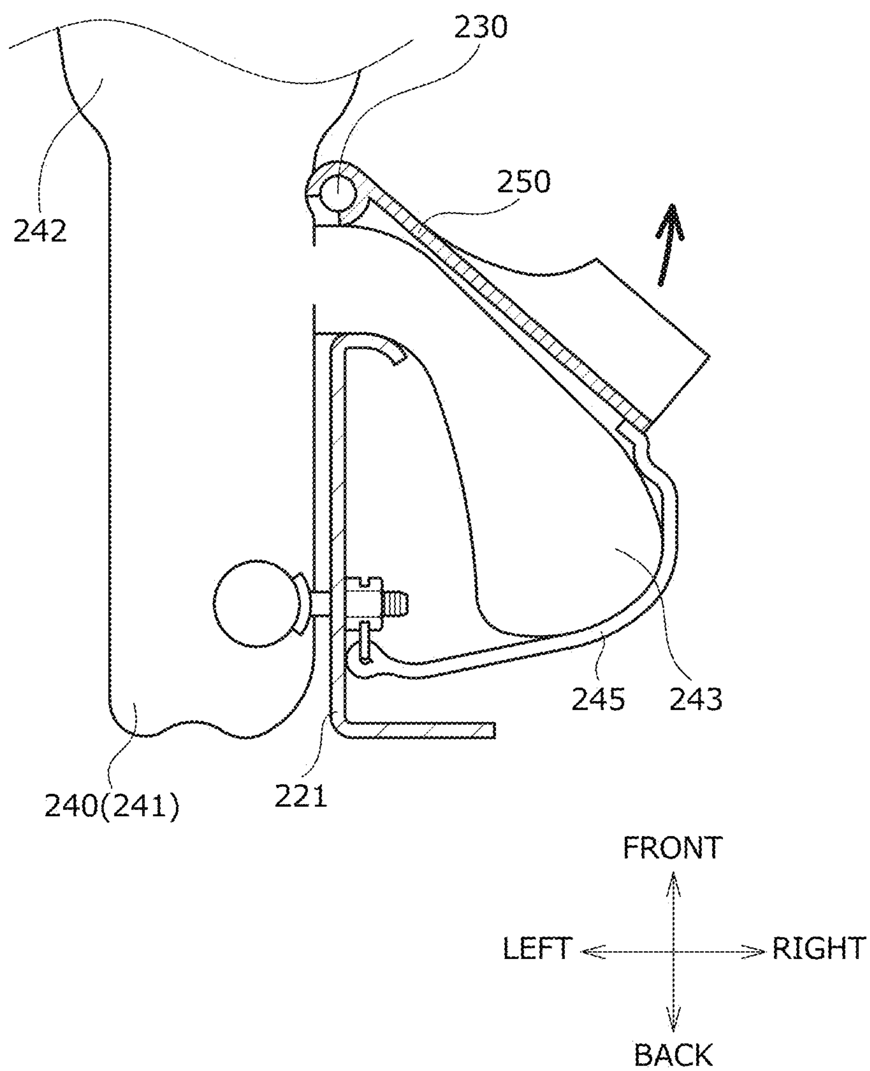


FIG. 12

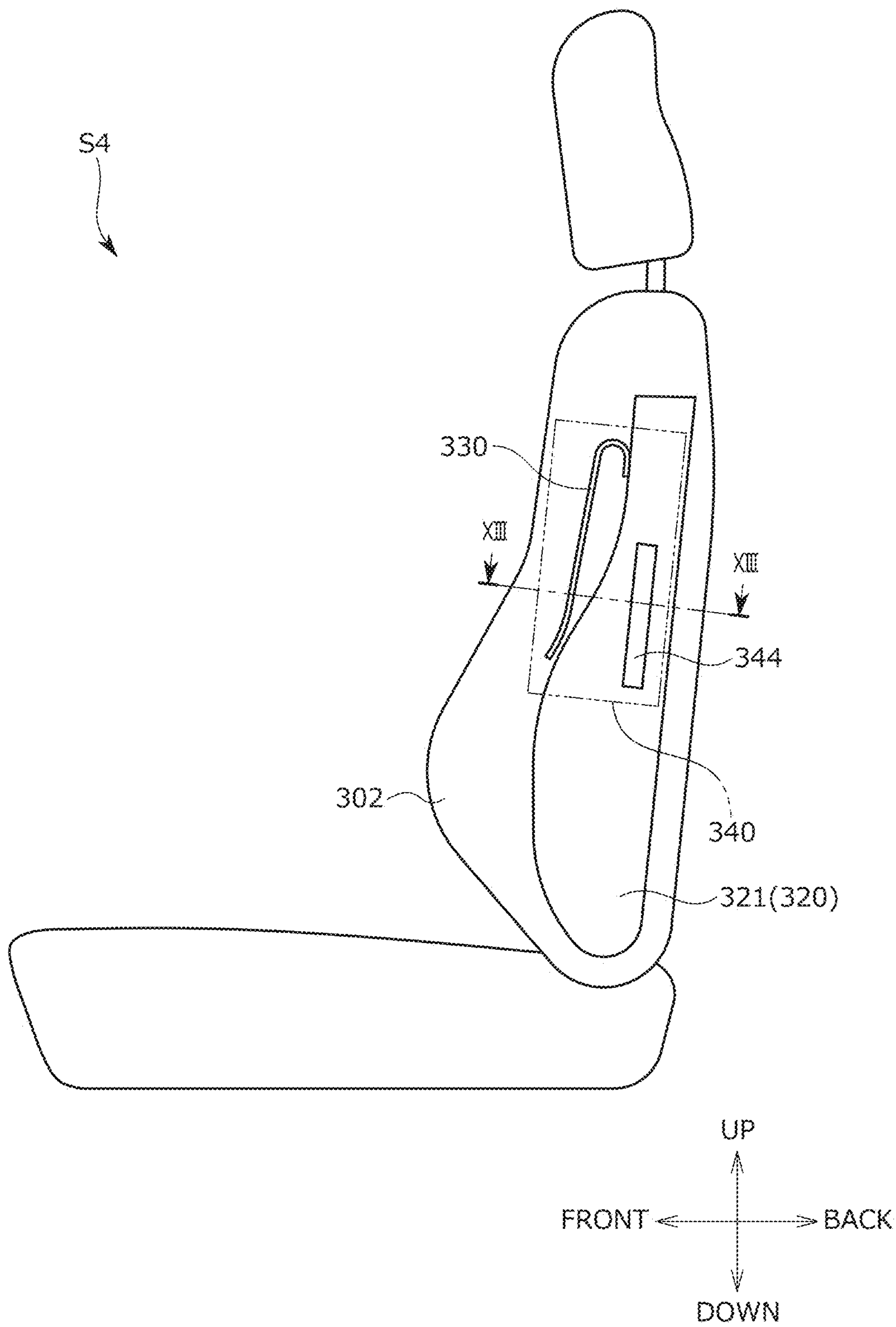


FIG. 13

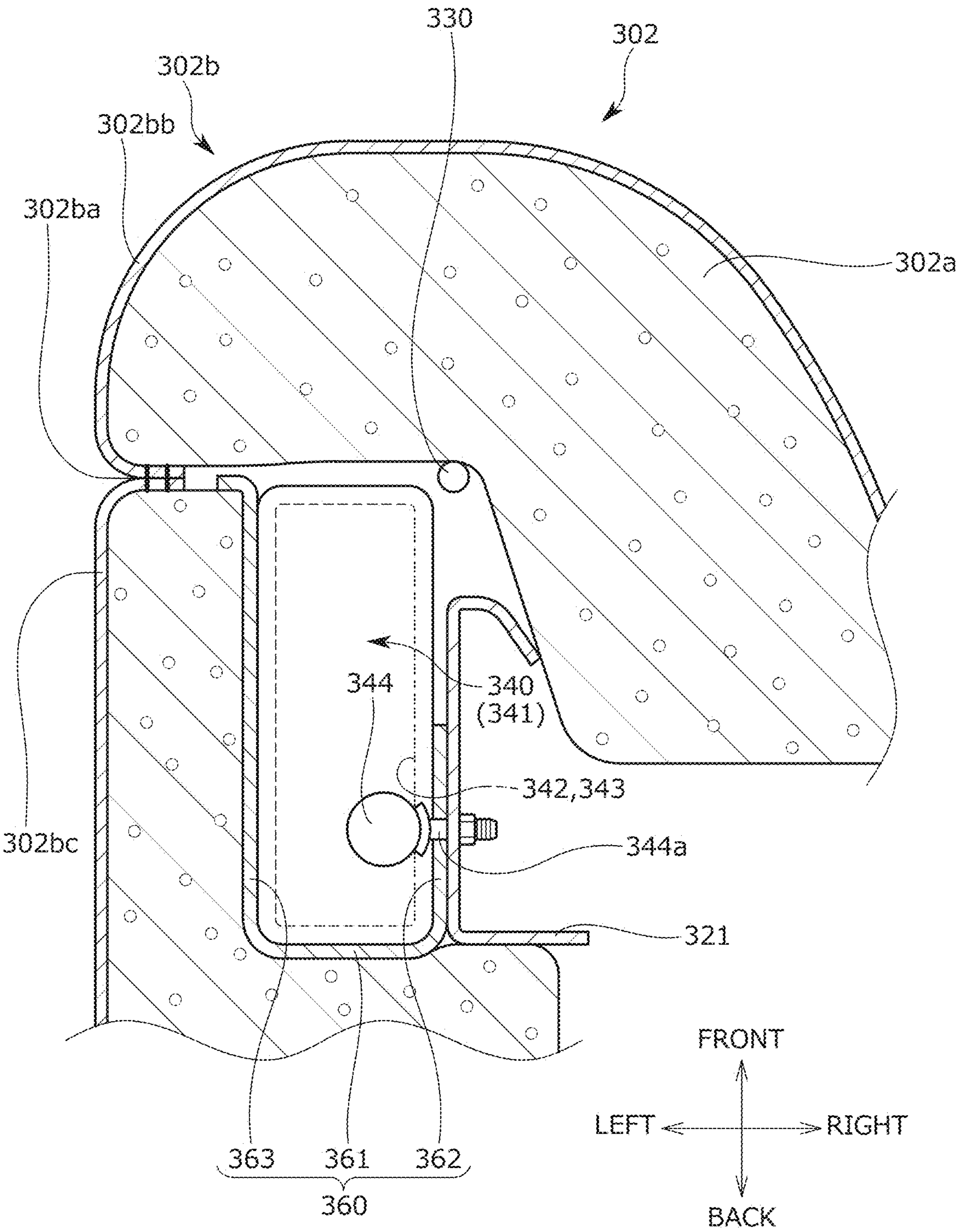

330
302
302b
302bb
302ba
302a
344
340
(341)
342,343
302bc
344a
321
FRONT
LEFT
RIGHT
BACK
363
361
362
360

FIG. 14

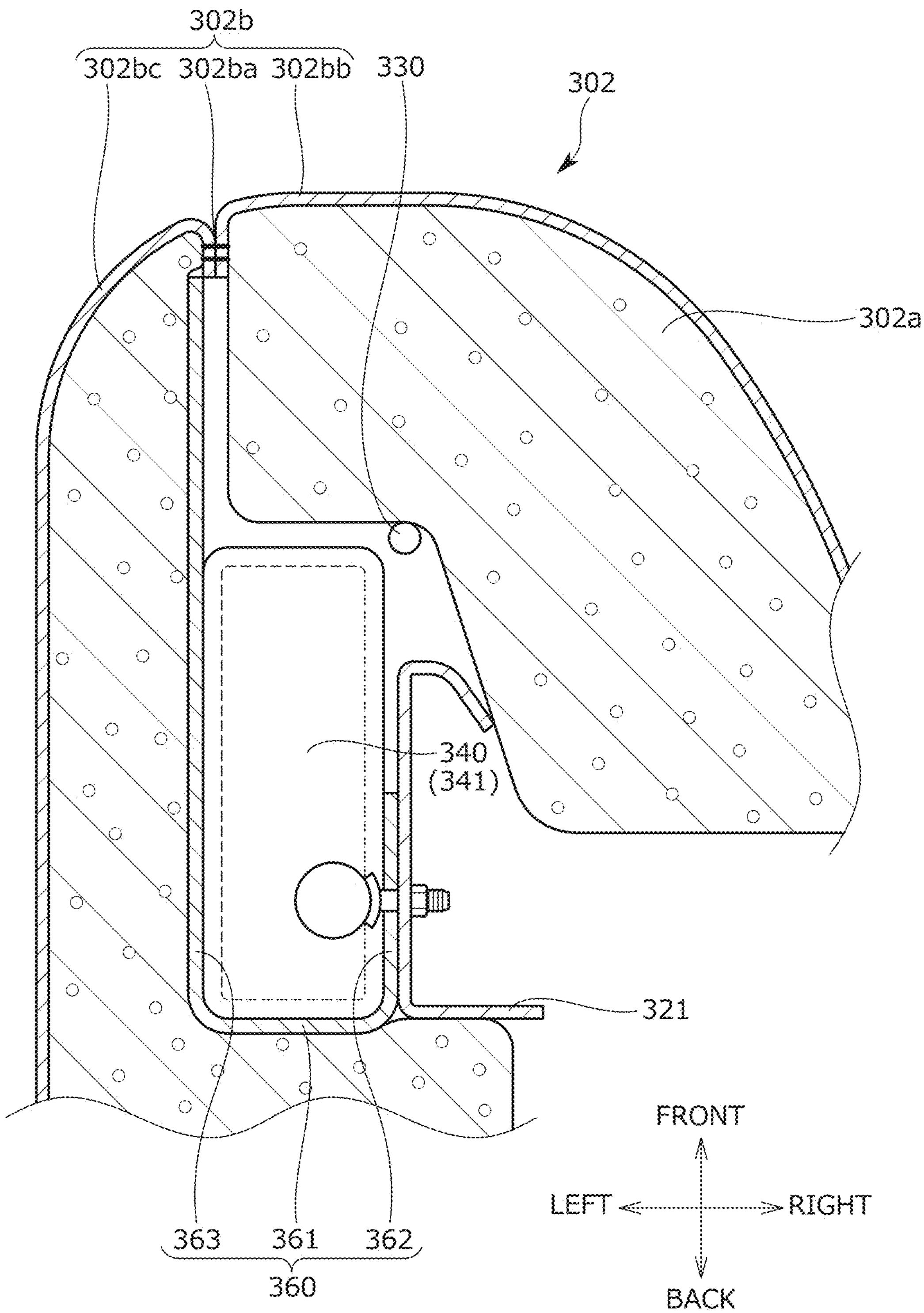

302b
302bc
302ba
302bb
330
302
302a
340
(341)
321
363
361
362
360
FRONT
LEFT
RIGHT
BACK

FIG. 15

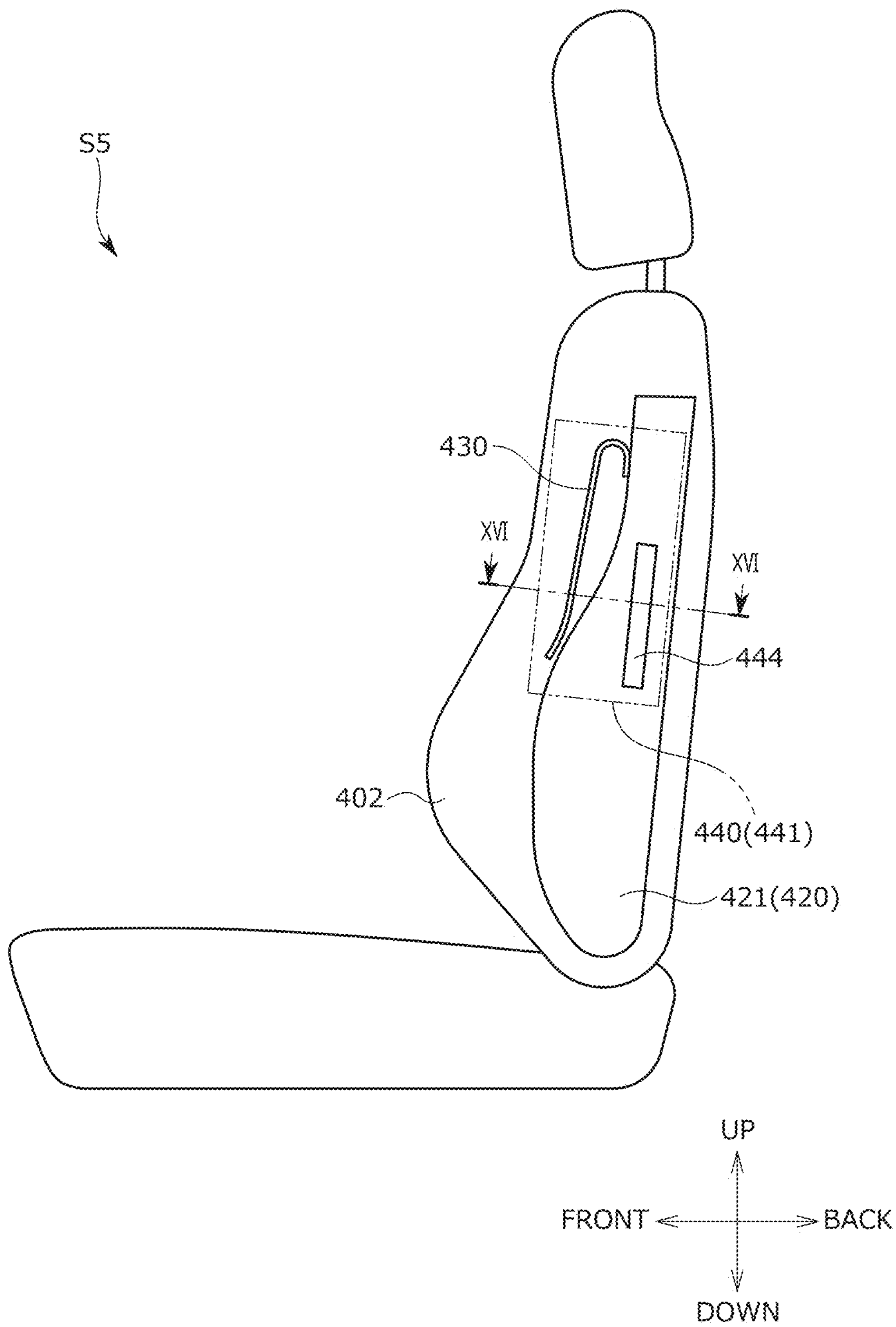


FIG. 16

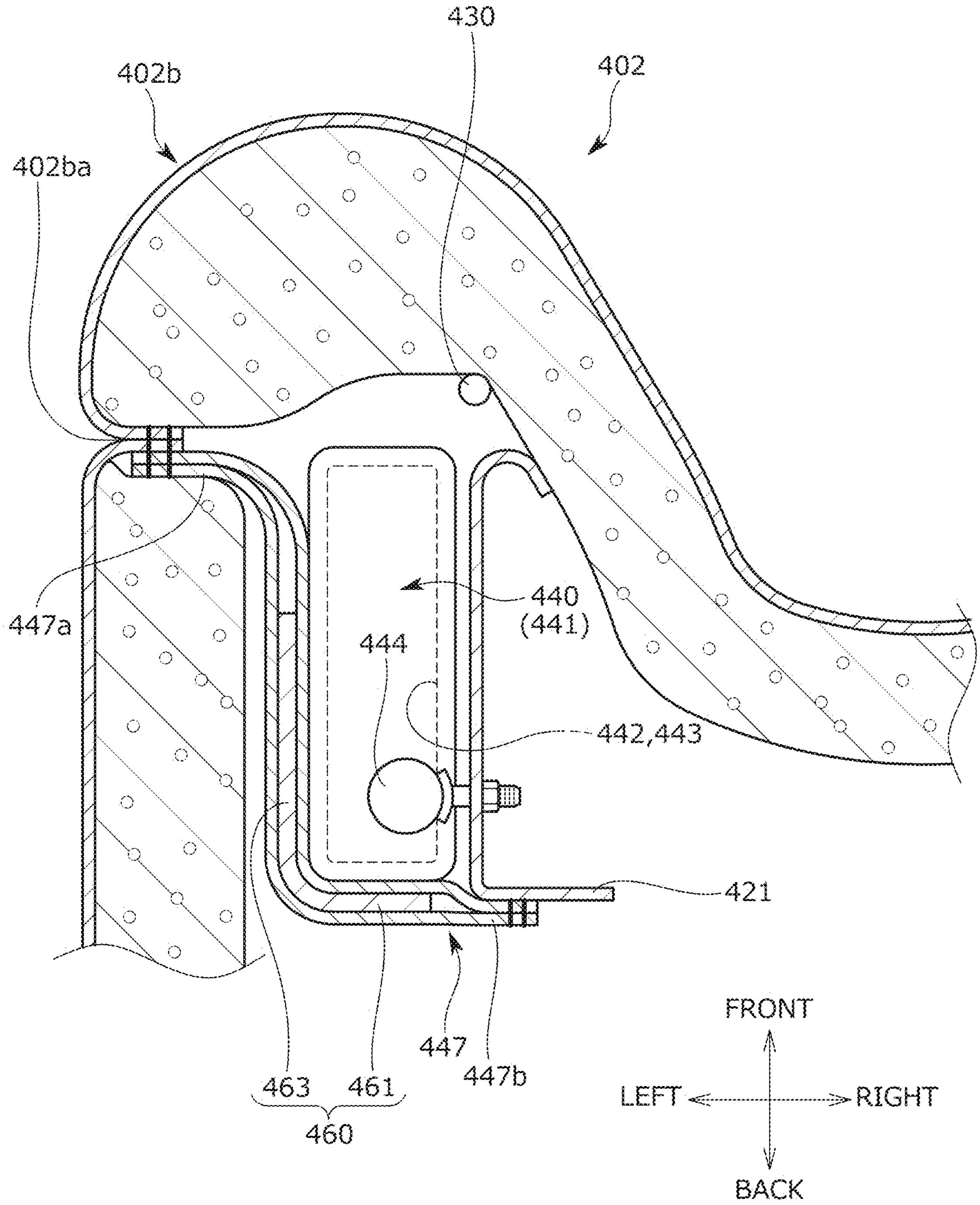

430
402b
402
402ba
447a
440
(441)
444
442,443
421
447
447b
463
461
460
FRONT
LEFT
RIGHT
BACK

FIG. 17

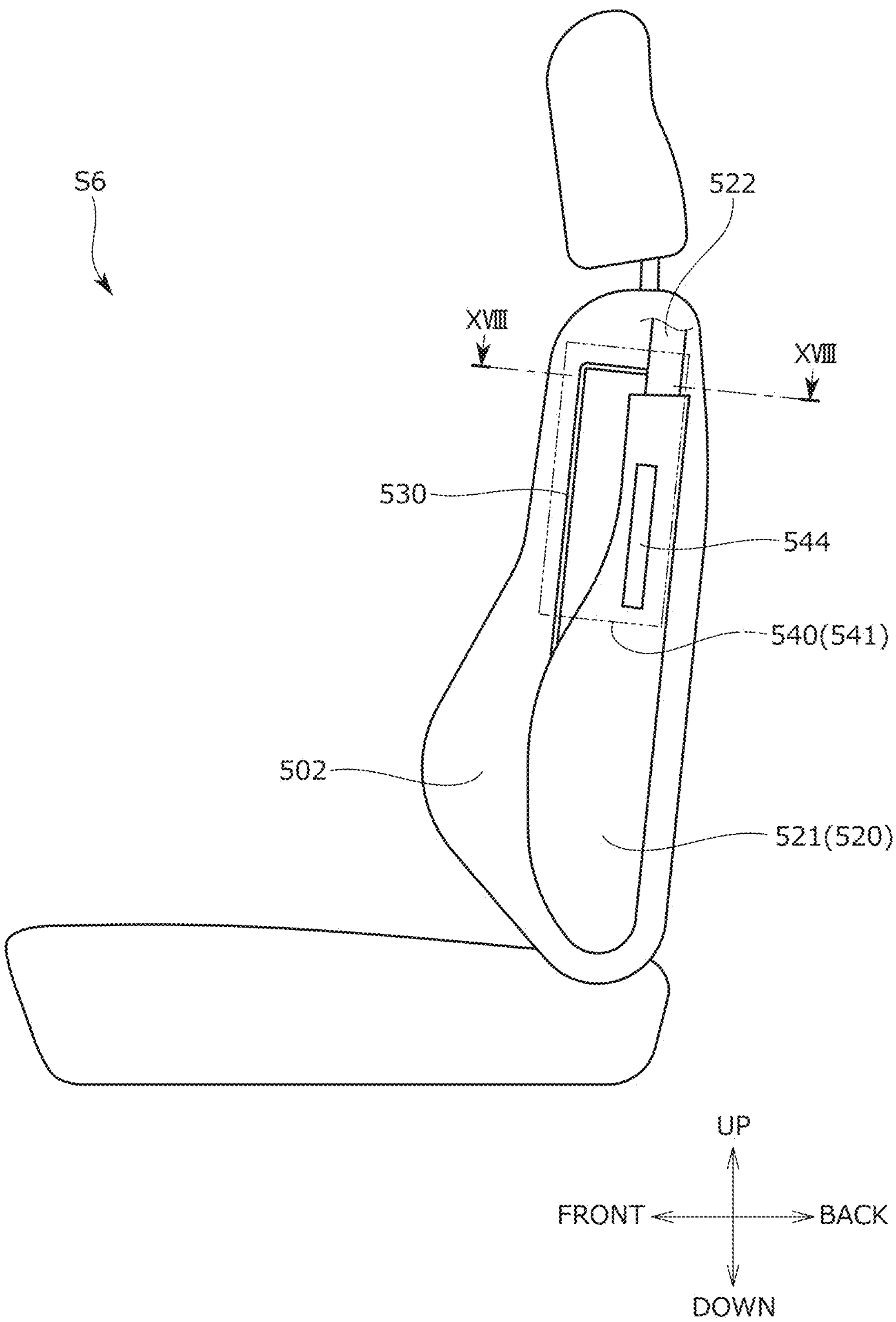

S6
522
XVIII
XVIII
530
544
540(541)
502
521(520)
UP
FRONT
BACK
DOWN

FIG. 18

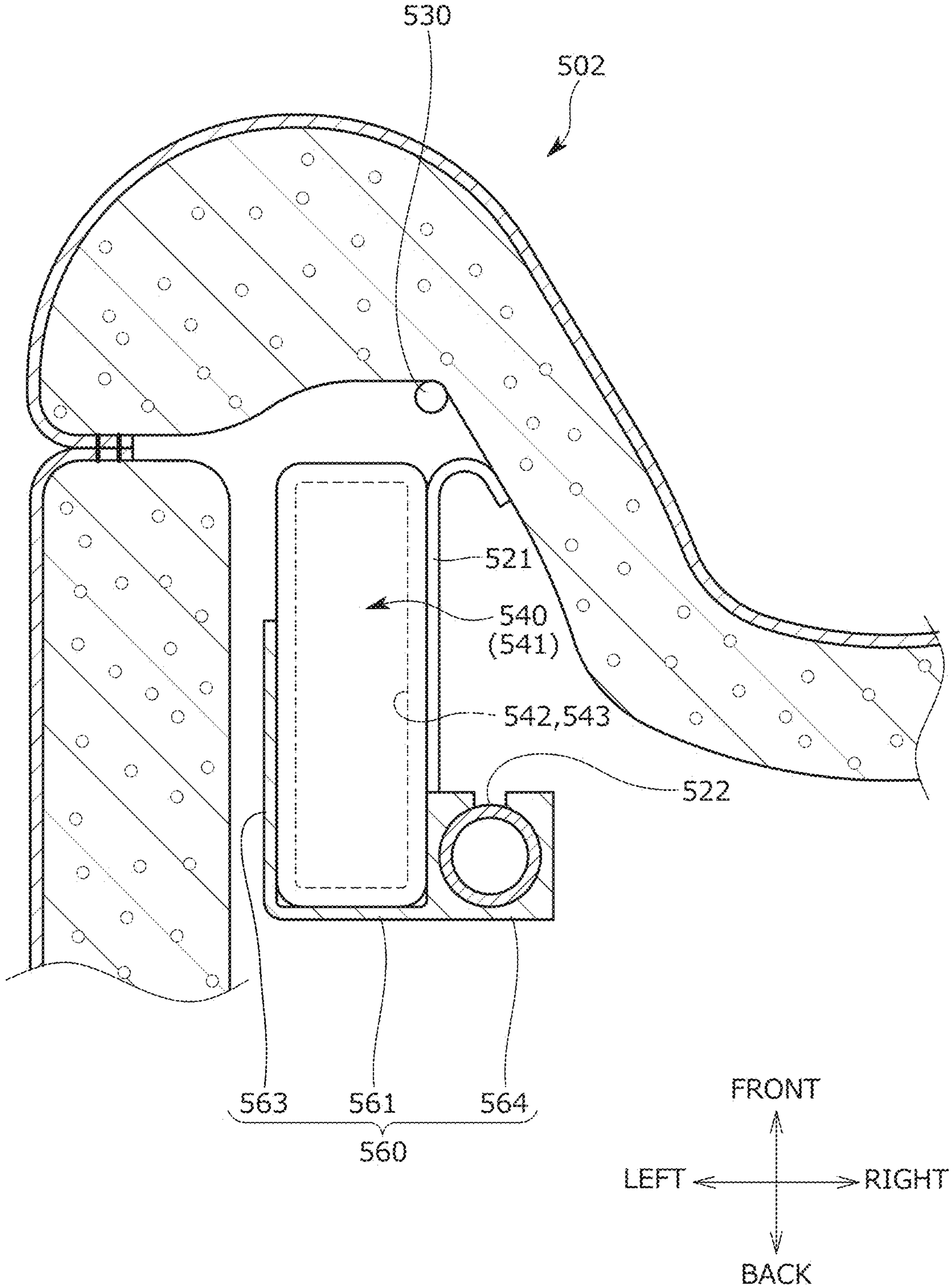

530
502
521
540
(541)
542,543
522
563
561
564
560
FRONT
LEFT
RIGHT
BACK

FIG. 19

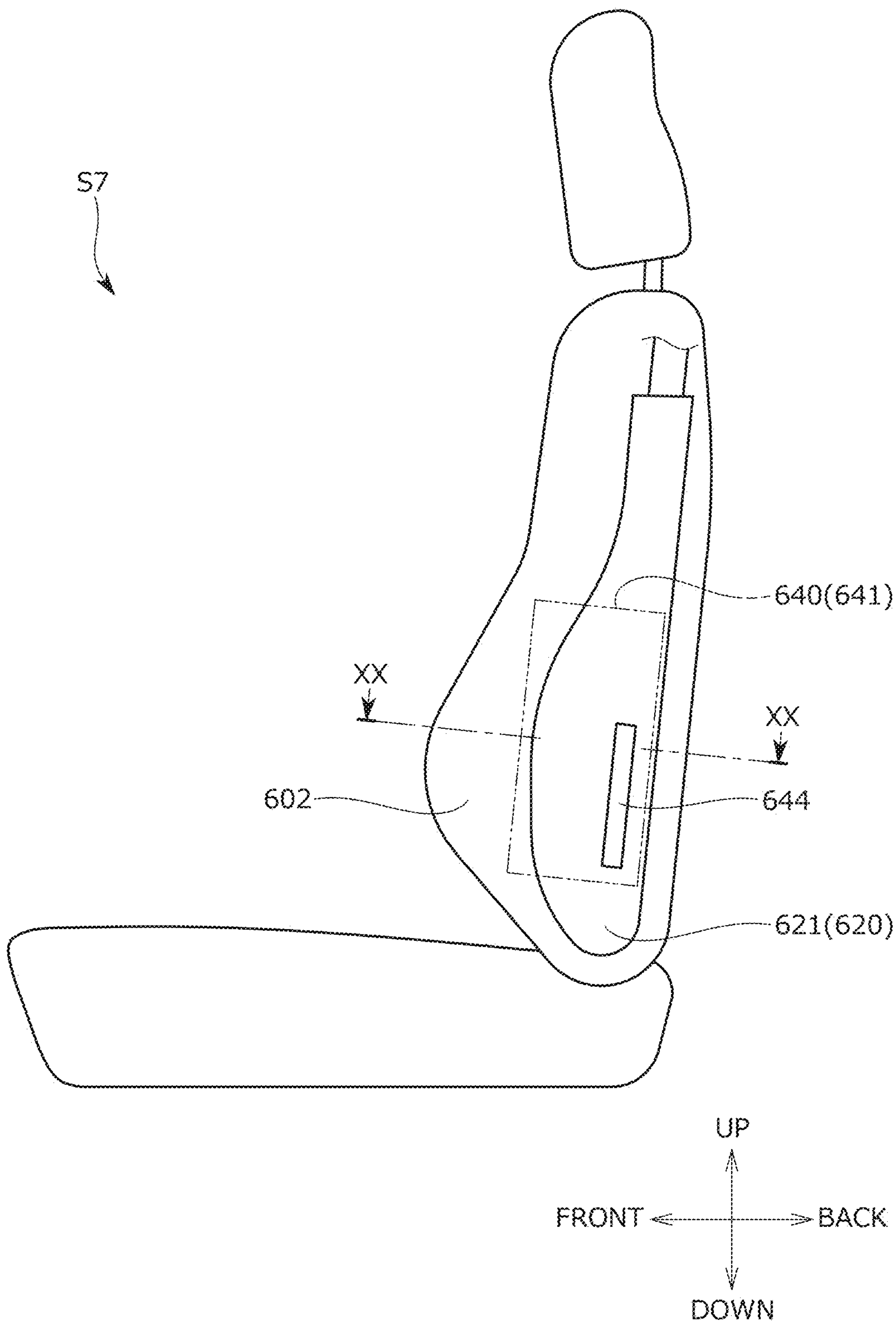


FIG. 20

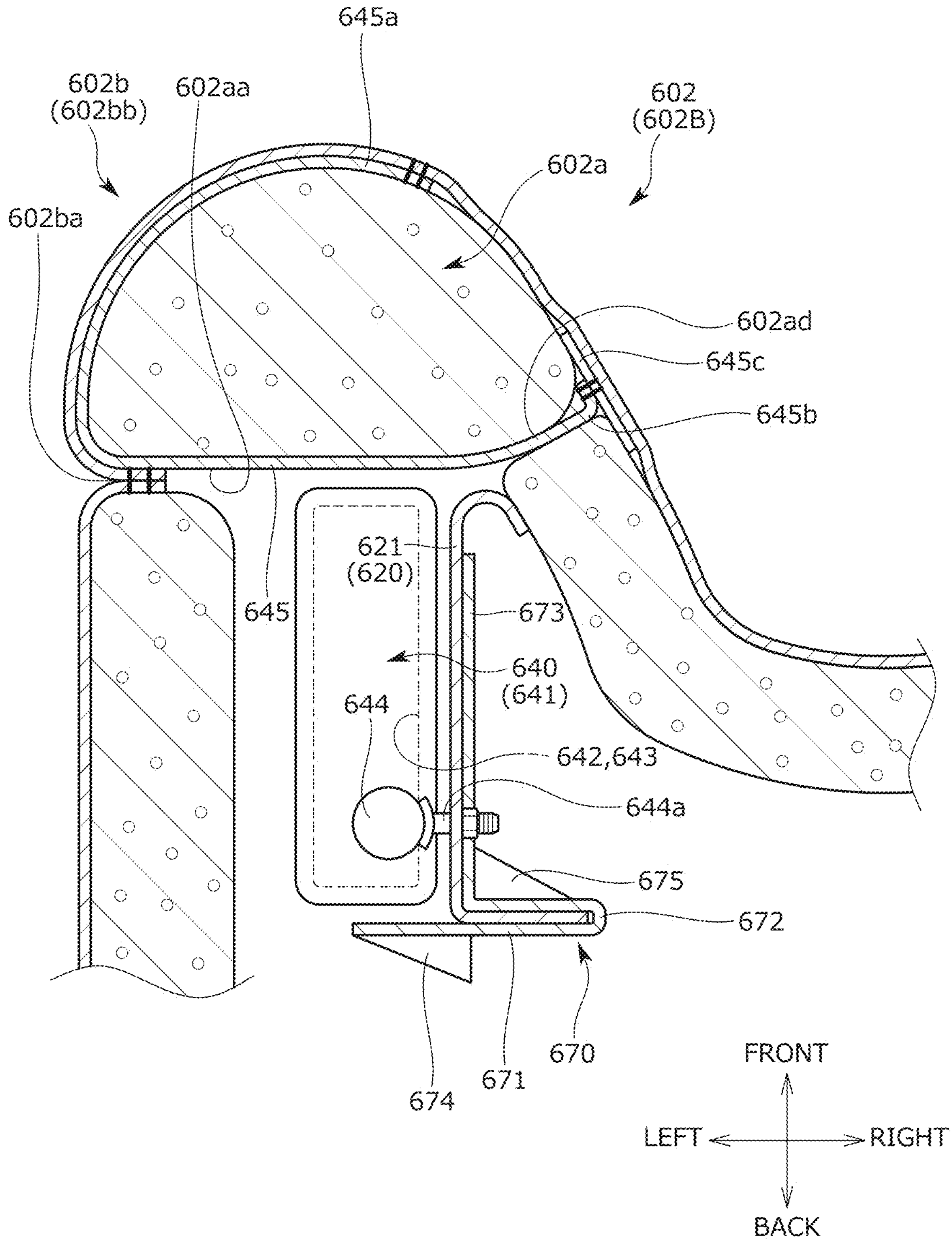
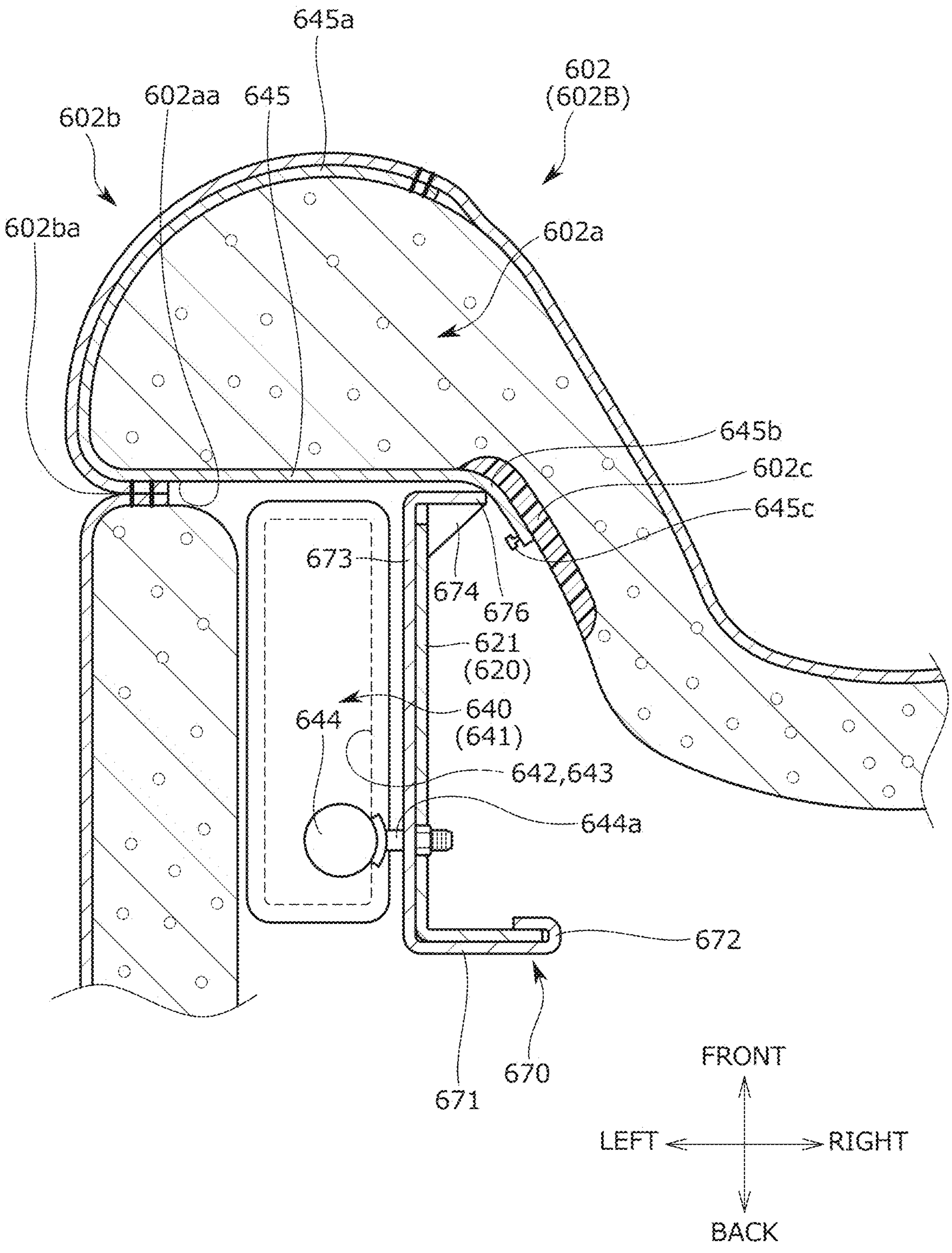

645a
602b
(602bb)
602aa
602
(602B)
602a
602ba
602ad
645c
645b
621
(620)
645
673
640
(641)
644
642,643
644a
675
672
674
671
670
FRONT
LEFT
RIGHT
BACK

FIG. 21


645a
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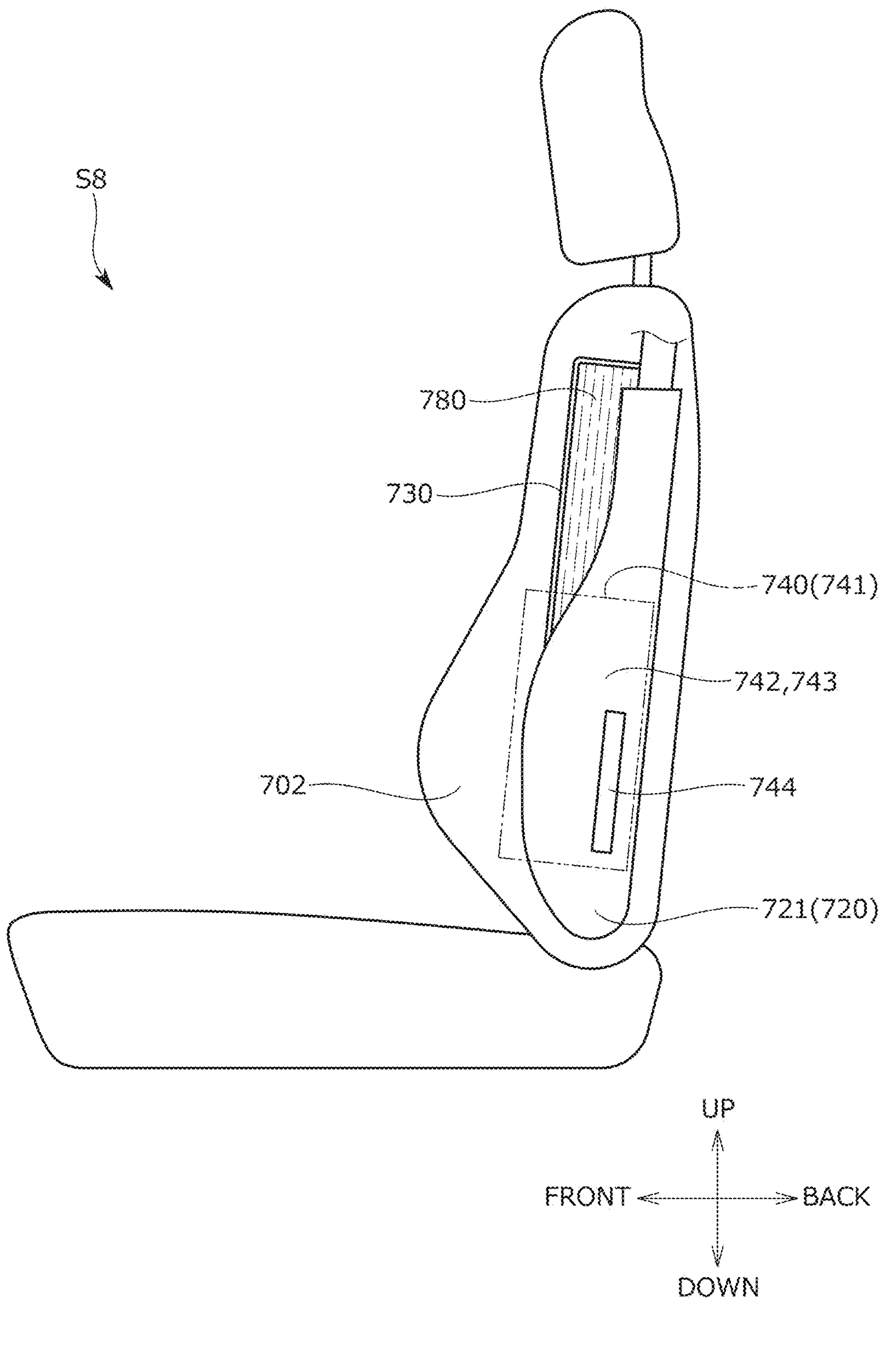
(602B)
602b
602aa
645
602ba
602a
645b
602c
645c
673
674
676
621
(620)
640
(641)
644
642,643
644a
672
670
671
FRONT
LEFT
RIGHT
BACK

FIG. 22



CONVEYANCE SEAT

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent Ser. No. 18/648,531 filed on Apr. 29, 2024, which, in turn, is a continuation of U.S. patent Ser. No. 18/024,985 (now U.S. Pat. No. 11,999,275) filed on Mar. 7, 2023, which, in turn, is a National Entry application of PCT Application Serial Number PCT/JP2021/033002, filed Sep. 8, 2021. Further, this application claims priority from U.S. Provisional Application No. 63/075,944, filed Sep. 9, 2020, and U.S. Provisional Application No. 63/082,066, filed Sep. 23, 2020, the entire contents of which are hereby incorporated by reference into this application.

TECHNICAL FIELD

[0002] The present invention relates to a conveyance seat. More particularly, the present: invention relates to a conveyance seat including a side airbag device for mitigating an impact applied from a side of a conveyance.

BACKGROUND ART

[0003] Known in the related art is a vehicle seat including a side airbag device attached to a side part of the seat back in the seat width direction in order to mitigate an impact applied from a side of the vehicle (see, for example, PATENT LITERATURE 1).

[0004] The vehicle seat described in PATENT LITERATURE 1 includes a seat back and a side airbag device attached to a side portion of the seat back.

[0005] The seat back has a back frame having side frames disposed on the right and left sides in the seat width direction. The side airbag device has an airbag module attached to the outside surface of the side frame.

[0006] Further, the airbag module has a first airbag inflation-deployed on the outside surface side of the side frame, a second airbag inflation-deployed on the inside surface side of the side frame, and an inflator supplying gas into the first airbag and the second airbag.

[0007] By having the first airbag and the second airbag as described above, an occupant (occupant seated in the vehicle seat) can be quickly restrained in the initial stage of the airbag inflation deployment.

CITATION LIST

Patent Literature

[0008] PATENT LITERATURE 1: WO 2019/138954 A

SUMMARY OF INVENTION

Technical Problem

[0009] By the way, as for a vehicle seat including a side airbag device as in PATENT LITERATURE 1, it has been required to restrain a seated occupant more quickly and more efficiently by airbag inflation deployment in a more stable state.

[0010] Specifically, it has been required to inflation-deploy a second airbag in particular in a more stable state in a vehicle seat including a side airbag device having a first airbag inflation-deployed on the outside surface side of a side portion of the seat back and the second airbag inflation-deployed on the inside surface side of the side portion.

[0011] The present invention has been made in view of the above problems, and an object of the present invention is to provide a conveyance seat including a side airbag device and enabling airbag inflation deployment in a more stable state.

Solution to Problem

[0012] The above object is achieved by means of a conveyance seat including: a seat back serving as a backrest portion; and a side airbag device attached to a side part of the seat back in a seat width direction in order to mitigate an impact applied from a side of a conveyance, in which the seat back includes a back frame having side frames disposed on right and left sides in the seat width direction, and a side support member attached to a seat front part of the side frame and protruding to a seat front side beyond the side frame in order to cause a side portion of the seat back to protrude to the seat front side, the side support member elongatedly extends in an up to down direction along the side frame and is disposed to form a gap with the side frame in a seat front to back direction, the side airbag device includes an airbag module attached to one side surface that is either an outside surface or an inside surface of the side frame, the airbag module has a first airbag inflation-deployed on the one side surface side of the side frame, a second airbag inflation-deployed on the other side surface side of the side frame, and an inflator supplying gas into the first airbag and the second airbag, and the second airbag passes through the gap formed between the side frame and the side support member in the seat front to back direction and is inflation-deployed on the other side surface side of the side frame.

[0013] With the configuration described above, it is possible to realize a conveyance seat with which an airbag can be inflation-deployed in a more stable state.

[0014] Specifically, by further providing the side support member, the second airbag is capable of passing through the gap formed between the side frame and the side support member in the seat front to back direction and can be inflation-deployed on the other side surface side of the side frame. By doing so, the second airbag can be inflation-deployed in a more stable state. As a result, a seated occupant can be restrained quickly and efficiently.

[0015] At this time, the side support member may be a linear member attached to each of an upper part of the side frame and a lower part of the side frame and elongatedly extending in the up to down direction, the airbag module may be attached to the outside surface of the side frame, the first airbag may be inflation-deployed toward the seat front on the outside surface side of the side frame, and the second airbag may pass through the gap formed between the side frame and the linear member and be inflation-deployed toward an inside in the seat width direction on the inside surface side of the side frame.

[0016] With the configuration described above, the second airbag can be inflation-deployed in a stable state toward the inside in the seat width direction on the inside surface side of the side frame. As a result, the seated occupant can be restrained more quickly and more efficiently.

[0017] At this time, the seat back may include a protective member attached to the side surface of the side frame and covering the side frame, and the protective member may be disposed so as to cover at least a front end portion of the side frame and a rear end portion of the side frame.

[0018] In addition, the side frame may have a frame main body portion extending in the seat front to back direction, a front flange portion protruding to an inside in the seat width direction continuously from a front end portion of the frame main body portion, and a rear flange portion protruding to the inside in the seat width direction continuously from a rear end portion of the frame main body portion, the protective member may be a flexible protective plate, and the protective plate may have a plate main body portion covering an inside surface of the frame main body portion, a plate front portion folded back in order to cover the front flange portion continuously from a front end portion of the plate main body portion, and a plate rear portion folded back in order to cover the rear flange portion continuously from a rear end portion of the plate main body portion.

[0019] In addition, the back frame may have an upper frame connecting upper end portions of the right and left side frames, and the seat back may include a second protective member covering a part of the upper frame connected to the upper end portion of the side frame.

[0020] By providing the protective member protecting the side frame (side frame end portion in particular) as described above, it is possible to suppress unintentional contact of the inflation-deployed second airbag with the side frame (side frame end portion). As a result, the second airbag can be inflation-deployed stably.

[0021] At this time, the seat back may include a pad material disposed on a front surface of the back frame, a skin material covering the back frame and the pad material, and a movable body provided at the side portion of the seat back, disposed between the side frame and the pad material in the seat front to back direction, and movable in order to cause the pad material to protrude to the seat front side at the side portion, and the movable body may move to the seat front side as the second airbag is inflation-deployed and cause the pad material to protrude to the seat front side.

[0022] In addition, the movable body may have a rotating member attached to the side support member so as to be rotatable, rotating to the seat front side as the second airbag is inflation-deployed, and causing the pad material to protrude to the seat front side.

[0023] In addition, the rotating member may have a rotating main body portion attached to the side support member so as to be rotatable and extending to a seat rear from the side support member along a bottom surface of the pad material, and a protruding portion protruding from a front surface of the rotating main body portion toward the pad material and abutting against the pad material.

[0024] By providing the movable body (rotating body) movable to the seat front side as the second airbag is inflation-deployed as described above, the deployment force of the second airbag can be efficiently transmitted to the pad material. As a result, the seated occupant can be restrained more quickly and efficiently.

[0025] At this time, the side airbag device may further include a guide member guiding the inflation deployment of the second airbag, one end portion of the guide member may be attached to the side frame or an attachment bracket attached to the side frame, and the other end portion of the guide member may be attached to the movable body.

[0026] In addition, the side airbag device may further include a guide member guiding the inflation deployment of the second airbag, one end portion of the guide member may be attached to an attachment bracket attached to the side surface of the side frame, and the other end portion of the guide member may be attached to the second airbag, and the attachment bracket may be fastened together with the side frame by an assembly shaft provided on the inflator and protruding from the inflator toward the seat width direction.

[0027] In addition, the attachment bracket may have a slit disposed inside the side frame in the seat width direction for the one end portion of the guide member to be inserted therethrough and attached.

[0028] By providing the guide member as described above, it is possible to prevent the inflation-deployed second airbag from slipping into the seat rear beyond the side frame.

[0029] In addition, by providing the attachment bracket as described above, the guide member can be attached stably.

[0030] In addition, since the attachment bracket is fastened together with the side frame by the attachment shaft protruding from the inflator toward the seat width direction, the airbag module and the guide member can be attached stably.

[0031] In addition, since the attachment bracket has the slit, one end portion of the guide member can be attached with ease.

[0032] At this time, the seat back may include a pad material disposed on a front surface of the back frame and a skin material covering the back frame and the pad material, the side airbag device may have a retainer member holding the airbag module from a seat rear side, the retainer member may have a rear wall portion disposed on the seat rear side of the airbag module, and a side wall portion extending to the seat front side along a side surface of the airbag module continuously from one end portion of the rear wall portion in the seat width direction, and the side wall portion may extend to the seat front side up to a skin burst-open portion or a position reaching a vicinity of the skin burst-open portion provided at the skin material and burst-opening when the first airbag is inflation-deployed.

[0033] In addition, the side airbag device may have a retainer member holding the airbag module from a seat rear side, the retainer member may have a rear wall portion disposed on the seat rear side of the airbag module, a side wall portion disposed along the other side surface of the side frame, and a reinforcement portion attached to a rear surface of the rear wall portion and protruding to the seat rear, and the side wall portion may be fastened together with the side frame by an assembly shaft provided on the inflator and protruding from the inflator toward the seat width direction.

[0034] By having the retainer member as described above, the retainer member is capable of suitably receiving pressure entailed by the inflation deployment of the airbag. As a result, the inflation deployment directions of the first airbag and the second airbag can be stabilized.

[0035] In addition, the retainer member extends to the seat front side up to the skin burst-open portion or the position reaching the vicinity of the skin burst-open portion burst-opening when the first airbag is inflation-deployed. Therefore, the pressure received by the retainer member can be efficiently transmitted to the skin burst-open portion to burst-open the skin burst-open portion.

[0036] In addition, the retainer member can be stably attached since the retainer member is fastened together with the side frame by the assembly shaft protruding from the inflator toward the seat width direction.

[0037] At this time, the side airbag device may have a retainer member holding the airbag module from a seat lateral side, the retainer member may have a side wall portion disposed along the side surface of the side frame, a front wall portion protruding in the seat width direction continuously from a front end portion of the side wall portion and disposed on a seat rear side of the second airbag when the second airbag is inflation-deployed, and a reinforcement portion attached to a rear surface of the front wall portion and protruding to the seat rear, and the side wall portion may be fastened together with the side frame by an assembly shaft provided on the inflator and protruding from the inflator toward the seat width direction.

[0038] By having the retainer member as described above, the retainer member is capable of suitably receiving pressure entailed by the inflation deployment of the second airbag. As a result, the inflation deployment direction of the second airbag can be stabilized.

[0039] In addition, since the retainer member has the reinforcement portion, the holding force of the retainer member can be increased efficiently.

[0040] At this time, the seat back may include a pad material disposed on a front surface of the back frame and a skin material covering the back frame and the pad material, the pad material may have a pad through hole formed at a position in front of the airbag module in the side portion of the seat back and penetrating the pad material from an outside surface to an inside surface, the skin material may have a skin burst-open portion burst-opening during the inflation deployment of the first airbag on an outside surface of the side portion, the side airbag device may include a guide member guiding the inflation deployment of the first airbag, and the guide member may have a configuration in which one end portion of the guide member is attached to a front part or an outside part of the skin material at the side portion, the guide member extends continuously from the one end portion of the guide member and passes through the pad through hole formed in the outside surface of the pad material at the side portion, and the other end portion of the guide member passes through the pad through hole formed in the inside surface of the pad material at the side portion and is attached to an inside part of the skin material at the side portion.

[0041] With the above configuration, the guide member is capable of suitably guiding the pad material moving along with the inflation deployment of the airbag on the side portion of the seat back. As a result, the airbag can be inflation-deployed more quickly.

Advantageous Effects of Invention

[0042] According to the present invention, it is possible to realize a conveyance seat with which an airbag can be inflation-deployed in a more stable state. Further, an occupant seated in the conveyance seat can be restrained quickly and efficiently.

[0043] In addition, according to the present invention, the second airbag can be inflation-deployed in a stable state toward the inside in the seat width direction on the inside surface side of the side frame.

[0044] In addition, according to the present invention, by means of the protective member, it is possible to suppress contact of the inflation-deployed second airbag with the side frame (side frame end portion).

[0045] In addition, according to the present invention, by means of the movable body, the deployment force of the second airbag can be efficiently transmitted to the pad material. As a result, the seated occupant can be restrained more quickly and efficiently.

[0046] In addition, according to the present invention, by means of the guide member, it is possible to prevent the inflation-deployed second airbag from slipping into the seat rear beyond the side frame.

[0047] In addition, according to the present invention, by means of the attachment bracket, the guide member and the airbag module can be attached stably. In addition, one end portion of the guide member can be attached with ease.

[0048] In addition, according to the present invention, by means of the retainer member, pressure entailed by the inflation deployment of the airbag can be received suitably. In addition, the pressure received by the retainer member can be transmitted to the skin burst-open portion to efficiently burst-open the skin burst-open portion. In addition, the retainer member can be attached stably.

[0049] In addition, according to the present invention, by means of the retainer member, pressure entailed by the inflation deployment of the second airbag can be received suitably. In addition, the holding force of the retainer member can be efficiently increased by the reinforcement portion.

[0050] In addition, according to the present invention, by means of the guide member, it is possible to suitably guide the pad material moving along with the inflation deployment of the airbag on the side portion of the seat back.

BRIEF DESCRIPTION OF DRAWINGS

[0051] FIG. **1** is an external perspective view of a conveyance seat of the present embodiment.

[0052] FIG. **2** is a perspective view illustrating a seat frame serving as a skeleton.

[0053] FIG. **3**A is a side view illustrating a back frame, a side support member, and a side airbag device.

[0054] FIG. **3**B is a side view illustrating a state where the airbag illustrated in FIG. **3**A is inflation-deployed.

[0055] FIG. **4**A is a sectional view taken along line IV-IV of FIG. **1**.

[0056] FIG. **4**B is a sectional view taken along line IV-IV of FIG. **1**, which illustrates a state where a first airbag and a second airbag are inflation-deployed.

[0057] FIG. **5** is a side view illustrating the back frame, the side support member, and a protective member.

[0058] FIG. **6** is an external side view of a conveyance seat of a second embodiment.

[0059] FIG. **7**A is a sectional view taken along line VII-VII of FIG. **6**.

[0060] FIG. **7**B is a sectional view taken along line VII-VII of FIG. **6**, which illustrates a state where the first airbag and the second airbag are inflation-deployed.

[0061] FIG. **8** is a perspective view in which the back frame is viewed from the inside in the seat width direction, which illustrates the back frame and an attachment bracket.

[0062] FIG. **9** is a view illustrating a modification example of the attachment bracket.

[0063] FIG. **10** is an external side view of a conveyance seat of a third embodiment.

[0064] FIG. **11**A is a sectional view taken along line XI-XI of FIG. **10**.

[0065] FIG. **11**B is a sectional view taken along line XI-XI of FIG. **10**, which illustrates a state where the first airbag and the second airbag are inflation-deployed.

[0066] FIG. **12** is an external side view of a conveyance seat of a fourth embodiment.

[0067] FIG. **13** is a sectional view taken along line XIII-XIII of FIG. **12**.

[0068] FIG. **14** is a sectional view taken along line XIII-XIII of FIG. **12**, which illustrates a modification example of a retainer member.

[0069] FIG. **15** is an external side view of a conveyance seat of a fifth embodiment.

[0070] FIG. **16** is a sectional view taken along line XVI-XVI of FIG. **15**.

[0071] FIG. **17** is an external side view of a conveyance seat of a sixth embodiment.

[0072] FIG. **18** is a sectional view taken along line XVIII-XVIII of FIG. **17**.

[0073] FIG. **19** is an external side view of a conveyance seat of a seventh embodiment.

[0074] FIG. **20** is a sectional view taken along line XX-XX of FIG. **19**.

[0075] FIG. **21** is a sectional view taken along line XX-XX of FIG. **19**, which illustrates modification examples of the retainer member and a guide member.

[0076] FIG. **22** is an external side view of a conveyance seat of an eighth embodiment.

DESCRIPTION OF EMBODIMENTS

[0077] Embodiments according to the present invention will be described below with reference to FIGS. **1** to **22**.

[0078] The present embodiment relates to the invention of a conveyance seat including a side airbag device at a side portion of a seat back. The seat back includes a back frame having a side frame and a side support member attached to the front part of the side frame. The side airbag device includes a first airbag inflation-deployed on the outside surface side of the side frame and a second airbag inflation-deployed on the inside surface side of the side frame. The second airbag passes through the gap formed between the side frame and the side support member in the seat front to back direction and is inflation-deployed on the inside surface side of the side frame.

[0079] It should be noted that the side where a seated occupant sits with respect to the seat back of the conveyance seat is the seat front side.

First Embodiment

[0080] First, a conveyance seat S**1** of a first embodiment will be described with reference to FIGS. **1** to **5**.

[0081] As illustrated in FIGS. **1** and **2**, the conveyance seat S**1** is a vehicle seat and includes a seat main body having a seat cushion **1**, a seat back **2**, and a headrest **3**, side support members **30** attached to the side parts of the seat back **2** in the seat width direction in order to cause the side portions of the seat back **2** to protrude to the seat front side and be held, and a side airbag device **40** attached to the side part of the seat back **2** in the seat width direction in order to mitigate an impact applied from the side of the vehicle.

[0082] As illustrated in FIG. **1**, the seat cushion **1** is a seating portion that supports a seated occupant from below and is configured by placing a pad material la on a cushion frame **10**, which serves as a skeleton and is illustrated in FIG. **2**, and being covered with a skin material **1***b*.

[0083] The seat cushion **1** is configured from a middle portion **1**A at the middle part in the seat width direction and right and left side portions **1**B (side bolster portions) outside the middle portion **1**A in the seat width direction.

[0084] As illustrated in FIG. **1**, the seat back **2** is a backrest portion that supports the back of the seated occupant from behind and is configured by placing a pad material **2***a* on a back frame **20**, which serves as a skeleton and is illustrated in FIG. **2**, and being covered with a skin material **2**b.

[0085] The seat back **2** is configured from a middle portion **2**A at the middle part in the seat width direction and right and left side portions **2**B outside the middle portion **2**A in the seat width direction.

[0086] It should be noted that right and left pull-in grooves extending in the up to down direction are formed on the surface of the pad material **2***a* so as to divide the middle portion **2**A and the right and left side portions **2**B.

[0087] As illustrated in FIG. **1**, the headrest **3** is a head portion that supports the head of the seated occupant from behind and is configured by placing a pad material **3***a* on a pillar **3***c* serving as a core material and being covered with a skin material **3***b*.

[0088] It should be noted that a pillar attachment member **26** for attaching the pillar **3***c* supporting the main body of the headrest **3** is assembled to the upper portion of the back frame **20**.

[0089] As illustrated in FIG. **2**, the cushion frame **10** as a rectangular frame-shaped body is configured mainly from cushion side frames **11** disposed on the right and left sides, a plate-shaped pan frame **12** (bridging frame) provided to serve as a bridge between the front end parts of the cushion side frames **11**, a rear connection frame **13** connecting the rear parts of the cushion side frames **11**, and a plurality of support members **14** (elastic springs) hooked on the pan frame **12** and the rear connection frame **13** and extending in the seat front to back direction.

[0090] The cushion side frame **11** is a plate-shaped frame elongated in the seat front to back direction.

[0091] It should be noted that a reclining device **4** is attached to the rear part of the cushion side frame **11** and a rail device **6** is attached to the lower part of the cushion side frame **11** via a height link device **5**.

[0092] As illustrated in FIG. **2**, the back frame **20** as a rectangular frame-shaped body is configured mainly from side frames **21** disposed on the right and left sides, an inverted U-shaped upper frame **22** interconnecting the upper end parts of the side frames **21**, a plate-shaped lower frame **23** interconnecting the lower end parts of the side frames **21**, a plurality of wire members **24** (elastic wires) respectively hooked on the side frames **21** and extending in the seat width direction, and a support plate **25** held by the plurality of wire members **24** and supporting the seated occupant.

[0093] It should be noted that the back frame **20** further includes the pillar attachment member **26** attached to the middle part of the upper frame **22** in the seat width direction in order to attach the pillar **3***c* of the headrest **3**.

[0094] The side frame **21** is a sheet metal member extending in up to down direction and having a substantially C-shaped cross section. The lower end part of the side frame **21** is connected to the rear end part of the cushion side frame **11** via the reclining device **4**.

[0095] By means of the reclining device **4**, the back frame **20** is capable of rotating relative to the cushion frame **10**.

[0096] Specifically, as illustrated in FIGS. **2** and **4**A, the side frame **21** has a frame main body portion **21***a* extending in the seat front to back direction, a front flange portion **21***b* protruding to the inside in the seat width direction continuously from the front end portion of the frame main body portion **21***a*, and a rear flange portion **21***c* protruding to the inside in the seat width direction continuously from the rear end portion of the frame main body portion **21***a*.

[0097] In addition, as illustrated in FIGS. **4**A and **4**B, a protective member **27** is attached to the inside surface of the side frame **21** to cover the side frame **21** from the inside in the seat width direction.

[0098] In addition, as illustrated in FIG. **5**, a second protective member **28** is attached to the lower end part of the upper frame **22** that is connected to the upper end portion of the side frame **21** and covers the lower end part from below.

[0099] The protective member **27** is a flexible protective plate that protects the side frame **21**.

[0100] As illustrated in FIGS. **4**A and **4**B, the protective member **27** has a plate main body portion **27***a* covering the inside surface of the frame main body portion **21***a*, a plate front portion **27***b* folded back in order to cover the front flange portion **21***b* continuously from the front end portion of the plate main body portion **27***a*, and a plate rear portion **27***c* folded back in order to cover the rear flange portion **21***c* continuously from the rear end portion of the plate main body portion **27***a*.

[0101] The second protective member **28** is a flexible protective cap that protects the upper frame **22** and is attached to the lower end portion of the pipe-shaped upper frame **22**.

[0102] The protective member **27** and the second protective member **28** have the function of preventing the airbag of the side airbag device **40** from coming into direct contact with the back frame **20** when the airbag has been inflation-deployed.

[0103] As illustrated in FIG. **5**, the protective member **27** is attached along the side frame **21** so as to be elongated in the up to down direction.

[0104] In addition, the protective member **27** is disposed at a position overlapping the side airbag device **40** in the up to down direction and the seat front to back direction (specifically, the same position).

[0105] As illustrated in FIGS. **2**, **3**A, and **3**B, the side support members **30** are linear members made of a metal material (specifically, wire members) and are attached by welding to the seat front parts of the right and left side frames **21**.

[0106] The side support member **30** is attached to the front surface of the side frame **21**, protrudes to the seat front side beyond the side frame **21**, and is a member for causing the side portion 2B of the seat back **2** to protrude to the seat front side.

[0107] It should be noted that although details will be described later, the side support member **30** has the function of causing the side portion of the seat back **2** to protrude to the seat front side and holding the side portion and a function for inflation-deploying the airbag of the side airbag device **40** in a stable state.

[0108] Specifically, the side support member **30** is configured mainly from a wire main body portion **31** elongatedly extending along the elongation direction of the side frame **21** and a wire extending portion **32** extending rearward from the upper end portion of the wire main body portion **31**.

[0109] Further, the wire extending portion **32** is welded and fixed in a state of abutting against the front surface of the upper part of the side frame **21**, and the lower end portion of the wire main body portion **31** is welded and fixed in a state of abutting against the front surface of the lower part of the side frame **21**.

[0110] In the above configuration, as illustrated in FIGS. **2** and **3**A, the side support member **30** elongatedly extends in the up to down direction along the side frame **21** and is disposed to form a gap G with the side frame **21** in the seat front to back direction.

[0111] In addition, the side support member **30** is disposed at a position overlapping the side airbag device **40** in the up to down direction and the seat front to back direction (specifically, the same position).

[0112] As illustrated in FIGS. **1** to **3**A and **3**B, the side airbag device **40** has a configuration in which folded airbags **42** and **43** are inflation-deployed, in a balloon shape and toward the seat front side, in the event of an impact from the side of the vehicle body.

[0113] As illustrated in FIGS. **4**A and **4**B, the side airbag device **40** includes an airbag module **41** provided in the pad material **2***a* in the side portion 2B of the seat back **2** and attached to the outside surface of the side frame **21**.

[0114] The airbag module **41** is configured mainly from the first airbag **42** inflation-deployed on the outside surface side of the side frame **21**, the second airbag **43** inflation-deployed on the inside surface side of the side frame **21**, an inflator **44** supplying gas into the first airbag **42** and the second airbag **43**, and a harness (not illustrated) connected to the inflator **44** and supplying electric power for ignition to the inflator **44**.

[0115] It should be noted that the harness is connected to a vehicular battery (not illustrated) disposed on the vehicle body.

[0116] As illustrated in FIGS. **4**A and **4**B, the first airbag **42** is inflation-deployed toward the seat front on the outside surface side of the side frame **21**.

[0117] The second airbag passes through the gap G formed between the side frame **21** and the side support member **30** and is inflation-deployed toward the seat front and the inside in the seat width direction on the inside surface side of the side frame **21**.

[0118] Specifically, the airbags **42** and **43** are inflation-deployed by gas supply from the inflator **44**, which is a gas generation source, toward the insides of the airbags **42** and **43**.

[0119] The inflator **44** is an elongated and substantially cylindrical gas generator and is disposed so as to be elongated in the up to down direction.

[0120] The inflator **44** has a plurality of assembly shafts **44***a* protruding to the inside in the seat width direction from the outside surface thereof and assembled to the side frame **21**.

[0121] It should be noted that the inflator **44** further has an airbag connecting portion (not illustrated) protruding from the outer surface thereof and connected to the inside of the airbag **11** and a harness connecting portion (not illustrated) formed on the outer surface of the inflator **44** and connected to the harness.

[0122] The plurality of assembly shafts **44***a* are disposed at intervals in the up to down direction. Each of the plurality of assembly shafts **44***a* passes through assembly holes provided in the side frame **21** and the protective member **27** and is fastened to an assembly nut **44***b*.

[0123] In other words, the assembly shaft **44***a* fastens the side frame **21** and the protective member **27** together with the assembly nut **44***b*.

[0124] In the above configuration, as illustrated in FIGS. **4**A and **4**B, in the event of inflation deployment of the airbags **42** and **43**, the pad material **2***a* is divided into an inner pad **2***ab* and an outer pad **2***ac* by a pad through hole **2***aa* serving as a branching point and deployed.

[0125] In addition, in the event of inflation deployment of the airbags **42** and **43**, the skin material **2***b* is divided into an inner skin material **2***bb* and an outer skin material **2***bc* by a skin burst-open portion **2***ba* serving as a branching point and deployed.

[0126] The pad through hole **2***aa* is a through hole penetrating the pad material **2***a* in the seat front to back direction and is formed so as to be elongated in the up to down direction.

[0127] Likewise, the skin burst-open portion **2***ba* is formed so as to be elongated in the up to down direction.

[0128] The pad through hole **2***aa* and the skin burst-open portion **2***ba* are formed at positions closer to the seat front than the airbag module **41** and are disposed at positions overlapping the airbag module **41** in the seat width direction (specifically, the same positions).

[0129] Therefore, the airbag can be inflation-deployed in a more stable state.

[0130] In addition, in the above configuration, as illustrated in FIGS. **4**A and **4**B, the second airbag **43** is configured to pass through the gap G formed between the side frame **21** and the side support member **30** in the seat front to back direction and be inflation-deployed on the inside surface side of the side frame **21**.

[0131] Therefore, the second airbag **43** can be inflation-deployed toward the seat front and the inside in the seat width direction in a more stable state.

[0132] In other words, by using the side support member **30** as a guide member, it is possible to suppress, for example, unintentional deployment of the second airbag **43** toward the seat front side.

[0133] In addition, in the above configuration, as illustrated in FIGS. **4**A, **4**B, and **5**, the protective member **27** and the second protective member **28** are attached so as to cover the side frame **21** and the upper frame **22**, respectively.

[0134] Regarding the protective member **27** in particular, the plate front portion **27***b* is provided so as to not only cover the tip portion of the side frame **21** (front flange portion **21***b*) so as to be folded back but also cover the entire (substantially the entire) front surface of the front flange portion **21***b*.

[0135] Further, the plate rear portion **27***c* also covers the tip portion of the side frame **21** (rear flange portion **21***c*) so as to be folded back. It should be noted that it is not particularly necessary to cover the entire rear surface of the side frame **21** and thus covering is performed up to the periphery of the tip portion of the rear flange portion **21***c*.

[0136] Therefore, unintentional contact of the inflation-deployed second airbag **43** with the end portion of the side frame **21** can be suitably suppressed.

Others

[0137] Although the side support member **30** and the side airbag device **40** in the above embodiment are attached to the side portion 2B of the seat back **2** as illustrated in FIGS. **1** and **2**, the attachment is not particularly limited and the side support member **30** and the side airbag device **40** may be attached to the side portion **1**B of the seat cushion **1**.

[0138] Although the side support member **30** in the above embodiment is a wire member formed of a metal material as illustrated in FIG. **2**, this is changeable and is not particularly limited.

[0139] The side support member **30** may be configured to hold the seated occupant from the side in the seat width direction and may be, for example, a plate-shaped member, a rod-shaped member, a movable plate, an inflatable bag member, or the like.

[0140] Although the protective member **27** in the above embodiment is configured as a protective plate as illustrated in FIGS. **4**A and **5**, this is changeable and is not particularly limited.

[0141] For example, the protective member **27** may be divided into a member covering the front end portion of the side frame **21** and a member covering the rear end portion of the side frame **21** and configured from the plurality.

[0142] Although the side airbag device **40** (airbag module **41**) in the above embodiment is attached to the outside surface of the side frame **21** as illustrated in FIGS. **4**A and **4**B, the attachment is not particularly limited and the side airbag device **40** (airbag module **41**) may be attached to the inside surface of the side frame **21**.

[0143] In the above case, it is preferable that the first airbag **42** is inflation-deployed on the inside surface side of the side frame and the second airbag **43** passes through the gap G formed between the side frame **21** and the side support member **30** and is inflation-deployed on the outside surface side of the side frame **21**.

Second Embodiment

[0144] Next, a conveyance seat S**2** of a second embodiment will be described with reference to FIGS. **6** to **9**.

[0145] It should be noted that description of content overlapping with the conveyance seat S**1** described above will be omitted.

[0146] As illustrated in FIGS. **6**, **7**A, and **7**B, the conveyance seat S**2** includes a seat main body having a seat back **102**, a side support member **130**, a side airbag device **140**, and a movable body **150** attached to a side portion **102**B of the seat back **102** and causing a pad material **102***a* of the side portion **102**B to protrude to the seat front side as the airbag is inflation-deployed.

[0147] As illustrated in FIGS. **7**A, **7**B, and **8**, the side airbag device **140** includes an airbag module **141** having a first airbag **142**, a second airbag **143**, and an inflator **144** and a guide member **145** guiding the inflation deployment of the second airbag **143**.

[0148] The guide member **145** is a cloth member (stay cloth) that guides the inflation deployment direction of the second airbag **143** to the seat front side and the inside in the seat width direction. The guide member **145** may be a strap.

[0149] The guide member **145** has a configuration in which one end portion **145***a* of the guide member **145** is attached to the front end portion of a side frame **121**, the guide member **145** extends to the seat rear continuously from the one end portion **145***a,* the guide member **145** is folded back near the rear end portion of the side frame **121**, and the other end portion **145***b* of the guide member is attached to the rear end portion of the movable body **150**.

[0150] Specifically, as illustrated in FIG. **8**, the one end portion **145***a* of the guide member **145** is attached to an attachment bracket **146** attached to the inside surface of the side frame **121**. In addition, as illustrated in FIGS. **7**A and **7**B, the other end portion **145***b* of the guide member **145** is attached to the back surface of the rear end portion of the movable body **150**.

[0151] Preferably, the other end portion **145***b* is attached to the back surface of the movable body **150** with an adhesive or the like. Alternatively, a trim cord (not illustrated) made of resin may be sewn to the other end portion **145***b* and the other end portion **145***b* may be hooked on the back surface of the movable body **150** via the trim cord.

[0152] It should be noted that the other end portion **145***b* may be attached to the tip portion of the second airbag **143**. In that case, it is preferable that the other end portion **145***b* is attached at the position that becomes the rear surface of the tip portion of the second airbag **143** when the second airbag **143** is inflation-deployed.

[0153] As illustrated in FIG. **8**, the attachment bracket **146** is a plate body having a substantially J-shaped cross section and is fastened together with the side frame **121** by an assembly shaft **144***a* protruding to the inside in the seat width direction from the inflator **144**.

[0154] In addition, the attachment bracket **146** has a slit **146***a* disposed inside the side frame **121** in the seat width direction for the one end portion **145***a* of the guide member **145** to be inserted therethrough and attached.

[0155] The airbag module **141** and the guide member **145** can be stably attached by the attachment bracket **146**. In addition, one end portion of the guide member **145** can be attached with ease.

[0156] It should be noted that the attachment bracket **146** may be made unnecessary, and the one end portion **145***a* of the guide member **145** may be directly attached to the side frame **121** with an adhesive or the like.

[0157] As illustrated in FIGS. **7**A and **7**B, the movable body **150** is a member that protrudes to the seat front side (from a normal position to a protruding position) as the second airbag **143** is inflation-deployed and is disposed between the side frame **121** and the pad material **102***a* in the seat front to back direction.

[0158] The movable body **150** has a rotating member **152** attached to the side support member **130** so as to be rotatable via an attachment portion **151**, rotating to the seat front side as the second airbag **143** is inflation-deployed, and causing the pad material **102***a* to protrude to the seat front side.

[0159] A plurality of the attachment portions **151**, which are hook-shaped members, are provided at intervals along the elongation direction of the side support member **130** to sandwich the side support member **130**.

[0160] The rotating member **152** has a rotating main body portion **152***a* rotatably attached to the side support member **130** and extending to the seat rear from the side support member **130** along the bottom surface of the pad material **102***a* and a plurality of protruding portions **152***b* (protruding ribs) protruding from the front surface of the rotating main body portion **152***a* toward the pad material **102***a* and abutting against the pad material **102***a*.

[0161] The rotating main body portion **152***a* is a plate body elongatedly extending in the up to down direction, and the plurality of protruding portions **152***b* are disposed at intervals along the elongation direction of the rotating main body portion **152***a*.

[0162] By providing the guide member **145** as described above, it is possible to prevent the inflation-deployed second airbag **143** from slipping into the seat rear beyond the side frame **121**.

[0163] In addition, by providing the movable body **150** as described above, the deployment force of the second airbag **143** can be efficiently transmitted to the pad material **102***a*. As a result, the seated occupant can be restrained more quickly and efficiently.

Others

[0164] In the above embodiment, as illustrated in FIGS. **7**A and **7**B, the movable body **150** protrudes to the seat front side as the second airbag **143** is inflation-deployed and efficiently transmits the deployment force of the second airbag **143** to the pad material **102***a*.

[0165] At this time, a pad high-hardness portion higher in hardness than the pad material **102***a* may be provided at the part of the back surface of the pad material **102***a* that faces the movable body **150** (rotating member **152**).

[0166] For example, the pad high-hardness portion may be molded integrally with the pad material **102***a* and may be at a part directly pressed by the movable body **150**.

[0167] By doing so, the movable body **150** is capable of more efficiently transmitting the deployment force of the second airbag **143** to the pad material **102***a*.

[0168] Although the attachment bracket **146** in the above embodiment is attached to the inside surface of the side frame **121** as illustrated in FIG. **8**, this is changeable and is not particularly limited.

[0169] For example, as illustrated in FIG. **9**, the attachment bracket **146** may be attached to the outside surface of the side frame **121**. Alternatively, the attachment bracket **146** may be attached to the front surface of the side frame **121**.

Third Embodiment of Conveyance Seat

[0170] Next, a conveyance seat S**3** of a third embodiment will be described with reference to FIGS. **10** to **11**A and **11**B.

[0171] It should be noted that description of content overlapping with the conveyance seats S**1** and S**2** described above will be omitted.

[0172] The conveyance seat S**3** includes a seat main body having a seat back **202**, a side support member **230**, a side airbag device **240**, and a movable body **250** attached to a side portion **202**B of the seat back **202** and causing a pad material **202***a* of the side portion **202**B to protrude to the seat front side as the airbag is inflation-deployed.

[0173] As illustrated in FIGS. **11**A and **11**B, the side airbag device **240** includes an airbag module **241** having a first airbag **242**, a second airbag **243**, and an inflator **244** and a guide member **245** guiding the inflation deployment of the second airbag **243**.

[0174] The airbag module **241** is attached to the outside surface of a side frame **221**. Specifically, an assembly shaft **244***a* of the inflator **244** penetrates the side frame **221** and is fastened with an assembly nut **244***b*.

[0175] The guide member **245** is a stay cloth that guides the inflation deployment direction of the second airbag **243**. The guide member **245** may be a strap.

[0176] A ring-shaped attachment member **246** is attached to one end portion **245***a* of the guide member **245**.

[0177] The ring-shaped attachment member **246** is inserted into an attachment groove **244***c* (attached portion) formed in the outer surface of the assembly nut **244***b* and connected to the assembly nut **244***b*.

[0178] With the above configuration, the one end portion **245***a* of the guide member **245** is connected to the assembly nut **244***b* via the attachment member **246**. In addition, the other end portion **245***b* of the guide member **245** is attached to the back surface of the movable body **250** with an adhesive or the like.

[0179] With the conveyance seat S3 described above, the airbags **242** and **243** can be inflation-deployed in a more stable state. In particular, the second airbag **243** can be inflation-deployed in a stable state on the inside surface side of the side frame **221**.

Fourth Embodiment of Conveyance Seat

[0180] Next, a conveyance seat S**4** of a fourth embodiment will be described with reference to FIGS. **12** to **14**.

[0181] It should be noted that description of content overlapping with the conveyance seats S1 to S3 described above will be omitted.

[0182] As illustrated in FIGS. **12** and **13**, the conveyance seat S**4** includes a seat main body having a seat back **302**, a side support member **330**, a side airbag device **340** having an airbag module **341**, and a retainer member **360** holding the airbag module **341** from the seat rear side.

[0183] The retainer member **360** is a holding member made of metal or resin and holding airbags **342** and **343** and an inflator **344** from the seat rear side and receives pressure entailed when the airbags **342** and **343** are inflation-deployed.

[0184] The retainer member **360** has a substantially U-shaped cross section and has a rear wall portion **361** disposed on the seat rear side of the airbag module **341** and an inside wall portion **362** and an outside wall portion **363** extending to the seat front side continuously from both end portions of the rear wall portion **361** in the seat width direction.

[0185] The inside wall portion **362** and the outside wall portion **363** respectively extend in the seat front to back direction along the side surfaces of the airbag module **341**.

[0186] The inside wall portion **362** is fastened together with a side frame **321** by an assembly shaft **344***a* protruding to the inside in the seat width direction from the inflator **344**.

[0187] The outside wall portion **363** extends to the seat front side up to a skin burst-open portion **302***ba* or a position reaching the vicinity of the skin burst-open portion **302***ba* provided at a side part of a skin material **302***b* and burst-opening when the first airbag **342** is inflation-deployed.

[0188] Specifically, the tip portion (front end portion) of the outside wall portion **363** is disposed at a position overlapping the skin burst-open portion **302***ba* in the seat front to back direction and the up to down direction (the same position).

[0189] It should be noted that the tip portion of the outside wall portion **363** may be attached to the skin burst-open portion **302***ba* with an adhesive or the like. Preferably, the tip portion of the outside wall portion **363** is attached to the skin burst-open portion **302***ba* of an outer skin material **302***bc*.

[0190] In addition, the tip portion of the outside wall portion **363** may be attached to the skin burst-open portion **302***ba* via an attachment member (attachment plate), which is not illustrated.

[0191] By providing the retainer member **360** as described above, it is possible to suitably receive pressure entailed by the inflation deployment of the airbags **342** and **343**. In addition, the pressure received by the retainer member **360** can be transmitted to the skin burst-open portion **302***ba* to efficiently burst-open the skin burst-open portion **302***ba*. In addition, the retainer member **360** can be stably attached.

Modification Example 1 of Retainer Member

[0192] Next, Modification Example 1 of the retainer member **360** will be described with reference to FIG. **14**.

[0193] The retainer member **360** of Modification Example 1 has a substantially U-shaped cross section and has the rear wall portion **361** disposed on the seat rear side of the airbag module **341** and the inside wall portion **362** and the outside wall portion **363** extending to the seat front side continuously from both end portions of the rear wall portion **361** in the seat width direction.

[0194] The inside wall portion **362** and the outside wall portion **363** respectively extend in the seat front to back direction along the side surfaces of the airbag module **341**.

[0195] The outside wall portion **363** extends to the seat front side up to the skin burst-open portion **302***ba* or a position reaching the vicinity of the skin burst-open portion **302***ba* provided at the front end part of the skin material **302***b* and burst-opening when the first airbag **342** is inflation-deployed.

[0196] Specifically, the tip portion (front end portion) of the outside wall portion **363** extends to the seat front beyond the side support member **330** and is disposed at a position overlapping the skin burst-open portion **302***ba* in the seat front to back direction and the up to down direction (the same position).

[0197] It should be noted that the tip portion of the outside wall portion **363** may be directly attached to the skin burst-open portion **302***ba* or may be attached to the skin burst-open portion **302***ba* via an attachment member (attachment plate), which is not illustrated.

[0198] By providing the retainer member **360** as described above, it is possible to suitably receive pressure entailed by the inflation deployment of the airbags **342** and **343**. In addition, the pressure received by the retainer member **360** can be transmitted to the skin burst-open portion **302***ba* to efficiently burst-open the skin burst-open portion **302***ba*.

Fifth Embodiment of Conveyance Seat

[0199] Next, a conveyance seat S**5** of a fifth embodiment will be described with reference to FIGS. **15** and **16**.

[0200] It should be noted that description of content overlapping with the conveyance seats S1 to S4 described above will be omitted.

[0201] The conveyance seat S5 includes a seat main body having a seat back **402**, a side support member **430**, a side airbag device **440** having an airbag module **441**, and a retainer member **460** holding the airbag module **441** from the seat rear side.

[0202] The retainer member **460** is a holding member holding airbags **442** and **443** and an inflator **444** from the seat rear side.

[0203] The retainer member **460** has a substantially L-shaped cross section and has a rear wall portion **461** disposed on the seat rear side of the airbag module **441** and an outside wall portion **463** extending to the seat front side continuously from one end portion of the rear wall portion **461** in the seat width direction.

[0204] The outside wall portion **463** extends in the seat front to back direction along a side surface of the airbag module **441**.

[0205] As illustrated in FIG. **16**, the retainer member **460** is attached in a state of being accommodated in a bag-shaped holder member **447**.

[0206] The holder member **447** is formed by, for example, pasting two skin materials together and sewing the materials into a bag shape.

[0207] One end portion **447***a* of the holder member **447** in the seat width direction is connected to a skin burst-open portion **402***ba* provided at a side part of a skin material **402***b* and burst-opening when the first airbag **442** is inflation-deployed. In other words, the holder member **447** is attached integrally with the skin material **402***b*.

[0208] It should be noted that the one end portion **447***a* may extend to the seat front side up to a position reaching the vicinity of the skin burst-open portion **402***ba* and be connected to a predetermined position on the back surface of the skin material **402***b*.

[0209] The other end portion **447***b* of the holder member **447** in the seat width direction is attached to the back surface of the rear end portion of a side frame **421** with an adhesive or the like.

[0210] It should be noted that the other end portion **447***b* may be attached to the rear surface of the side frame **421** via an attachment member (attachment plate), which is not illustrated.

[0211] By providing the retainer member **460** as described above, it is possible to suitably receive pressure entailed by the inflation deployment of the airbags **442** and **443**. In addition, the retainer member **460** can be stably attached by providing the holder member **447**.

Sixth Embodiment of Conveyance Seat

[0212] Next, a conveyance seat S6 of a sixth embodiment will be described with reference to FIGS. **17** and **18**. It should be noted that description of content overlapping with the conveyance seats S1 to S5 described above will be omitted.

[0213] The conveyance seat S6 includes a seat main body having a seat back **502**, a side support member **530**, a side airbag device **540** having an airbag module **541**, and a retainer member **560** holding the airbag module **541** from the seat rear side.

[0214] As illustrated in FIG. **17**, the airbag module **541** is attached to the upper part of the seat back **502** and disposed so as to straddle a side frame **521** and an upper frame **522** in the up to down direction.

[0215] As illustrated in FIG. **18**, the retainer member **560** is a holding member holding airbags **542** and **543** and an inflator **544** from the seat rear side, is disposed at a position above the side frame **521**, and is disposed at the same height position as the lower end part of the upper frame **522**.

[0216] The retainer member **560** has a substantially U-shaped cross section and has a rear wall portion **561** disposed on the seat rear side of the airbag module **541**, an outside wall portion **563** extending to the seat front side continuously from one end portion (outside end portion) of the rear wall portion **561** in the seat width direction, and a frame attachment portion **564** protruding to the seat front side from the other end portion (inside end portion) of the rear wall portion **561** in the seat width direction and attached to the upper frame **522**.

[0217] The frame attachment portion **564** is a hook member having a substantially C-shaped cross section and can be attached so as to sandwich the pipe-shaped upper frame **522** from the seat rear.

[0218] It should be noted that the frame attachment portion **564** is not limited to a hook shape and may have any shape insofar as the frame attachment portion **564** can be attached to the upper frame **522**. The frame attachment portion **564** may be attached to the upper frame **522** with an adhesive or the like.

[0219] By providing the retainer member **560** as described above, it is possible to suitably receive pressure entailed by the inflation deployment of the airbags **542** and **543**.

[0220] In addition, the retainer member **560** can be stably attached by providing the frame attachment portion **564**.

Seventh Embodiment of Conveyance Seat

[0221] Next, a conveyance seat S7 of a seventh embodiment will be described with reference to FIGS. **19** to **21**.

[0222] It should be noted that description of content overlapping with the conveyance seats S1 to S6 described above will be omitted.

[0223] The conveyance seat S7 includes a seat main body having a seat back **602**, a side airbag device **640** having an airbag module **641**, and a retainer member **670** holding the airbag module **641** from the seat rear side.

[0224] As illustrated in FIG. **20**, the seat back **602** includes a pad material **602***a* disposed on the front surface of a back frame **620** and a skin material **602***b* covering the back frame **620** and the pad material **602***a*.

[0225] The pad material **602***a* has pad through holes **602***aa* and **602***ad* formed at positions in front of the airbag module **541** in a side portion **602**B of the seat back **602** and penetrating the pad material **602***a* from the outside surface to the inside surface.

[0226] The skin material **602***b* has a skin burst-open portion **602***ba* burst-opening during the inflation deployment of airbags **642** and **643** on the outside surface of the side portion **602**B.

[0227] As illustrated in FIG. **20**, the side airbag device **640** includes the airbag module **641** having the first airbag **642**, the second airbag **643**, and an inflator **644** and a guide member **645** guiding the inflation deployment of the airbags **642** and **643**.

[0228] The airbag module **641** is attached to the outside surface of a side frame **621**. Specifically, an assembly shaft **644***a* of the inflator **644** penetrates the side frame **621** and a retainer member **660** and is fastened with an assembly nut.

[0229] It should be noted that the side airbag device **640** may not have the second airbag **643**.

[0230] The guide member **645** is a stay cloth that guides the inflation deployment directions of the airbags **642** and **643**. The guide member **645** may be a strap.

[0231] One end portion **645***a* of the guide member **645** is attached by sewing to the back surface of the front end portion of the skin material **602***b* (specifically, an inner skin material **602***bb*).

[0232] The guide member **645** extends from the one end portion **645***a* to the seat rear so as to cover the pad material **602***a* (specifically, the pad material **602***a* of the side portion **602**B) from the outside and further extends so as to pass through the pad through hole **602***aa* of the pad material **602***a*.

[0233] The other end portion **645***b* of the guide member **645** passes between the pad material **602***a* and the airbag module **641** in the seat front to back direction, extends toward the inside in the seat width direction, and passes through the pad through hole **602***ad*. Then, the other end portion **645***b* is attached to the back surface of the inside end portion in the seat width direction of the side portion **602**B of the skin material **602***b*.

[0234] A plate-shaped attachment member **645***c* (for example, a trim cord made of resin) is sewn to the other end portion **645***b* of the guide member **645**. Further, the other end portion **645***b* is hooked so as to be sandwiched between the pad material **602***a* and the skin material **602***b* via the attachment member **645***c*.

[0235] Specifically, the attachment member **645***c* is folded when the other end portion **645***b* of the guide member **645** is passed through the pad through hole **602***aa*. Further, in hooking the other end portion **645***b*, the attachment member **645***c* is raised (raised so as to form a T shape) and hooked in the space between the pad material **602***a* and the skin material **602***b*.

[0236] By providing the guide member **645** as described above, it is possible to suitably guide the deployment direction of the side portion **602**B of the seat back **602** when the airbag is inflation-deployed. As a result, the airbag can be deployed with speed.

[0237] In addition, the guide member **645** can be easily attached by providing the attachment member **645***c*.

[0238] As illustrated in FIG. **20**, the retainer member **670** is a holding member having a substantially T-shaped cross section and holding the airbag module **641** from the seat rear side and is attached to the side frame **621**.

[0239] The retainer member **670** has a rear wall portion **671** disposed on the seat rear side of the airbag module **641** and the side frame **621**, a folded-back wall portion **672** formed by folding back the right end portion of the rear wall portion **671** and extending to the outside in the seat width direction along the rear end portion (rear flange portion **621***c*) of the side frame **621**, and a side wall portion **673** bent from the extending end portion of the folded-back wall portion **672** and extending to the seat front along the inside surface of the side frame **621**.

[0240] The retainer member **670** is attached to the side frame **621** from the inside in the seat width direction and is attached so as to sandwich a part (rear flange portion **621***c*) of the side frame **621**.

[0241] In addition, the retainer member **670** (side wall portion **673**) is fastened together with the side frame **621** by the assembly shaft **644***a* provided on the inflator **644** and protruding from the inflator **644** toward the inside in the seat width direction.

[0242] The retainer member **670** further a has first reinforcement portion **674** attached to the rear surface of the rear wall portion **671** and protruding to the seat rear and a second reinforcement portion **675** attached to the connection part between the folded-back wall portion **672** and the side wall portion **673** and protruding from the side wall portion **673** to the inside in the seat width direction.

[0243] Each of the first reinforcement portion **674** and the second reinforcement portion **675** is a reinforcement rib (screen portion) having a substantially triangular section and disposed at a position corresponding to the airbag module **641**.

[0244] Specifically, the first reinforcement portion **674** is disposed at a position overlapping the airbag module **641** in the seat width direction and the up to down direction (the same position) and protrudes in a direction away from the airbag module **641**.

[0245] The second reinforcement portion **675** is disposed at a position overlapping the airbag module **641** in the seat front to back direction and the up to down direction (the same position) and protrudes in a direction away from the airbag module **641**.

[0246] Each of the first reinforcement portion **674** and the second reinforcement portion **675** is disposed at a position close to the inflator **644** and disposed so as to surround the inflator **644**.

[0247] By providing the retainer member **670** as described above, it is possible to suitably receive pressure entailed by the inflation deployment of the airbags **642** and **643**.

Modification Example 1 of Guide Member

[0248] Next, Modification Example 1 of the guide member **645** will be described with reference to FIG. **21**.

[0249] The guide member **645** is a stay cloth that guides the inflation deployment directions of the airbags **642** and **643**. The one end portion **645***a* of the guide member **645** is attached by sewing to the back surface of the front end portion of the skin material **602***b*.

[0250] The guide member **645** extends from the one end portion **645***a* to the seat rear so as to cover the pad material **602***a* from the outside and further extends so as to pass through the pad through hole **602***aa* of the pad material **602***a*.

[0251] The other end portion **645***b* of the guide member **645** passes between the pad material **602***a* and the airbag module **641** in the seat front to back direction and extends toward the inside in the seat width direction. Then, the other end portion **645***b* is attached to the back surface of the inside portion in the seat width direction of the side portion **602**B of the pad material **602***a*.

[0252] Specifically, the other end portion **645***b* of the guide member **645** is attached to the back surface of the pad material **602***a* by the attachment member **645***c* (specifically, an attachment pin).

[0253] At this time, a pad high-hardness portion **602***c* higher in hardness than the pad material **602***a* is provided at the part of the back surface of the pad material **602***a* that faces the other end portion **645***b*.

[0254] The pad high-hardness portion **602***c* is molded integrally with the pad material **602***a* on the back surface of the pad material **602***a*.

[0255] By providing the guide member **645** as described above, it is possible to suitably guide the deployment direction of the side portion **602**B of the seat back **602** when the airbag is inflation-deployed.

[0256] In addition, the guide member **645** can be easily attached by providing the attachment member **645***c* and the pad high-hardness portion **602***c*.

Modification Example 2 of Retainer Member

[0257] Next, Modification Example 2 of the retainer member **670** will be described with reference to FIG. **21**.

[0258] The retainer member **670** of Modification Example 2 is a holding member having a substantially C-shaped cross section and holding the airbag module **641** from a seat lateral side and is attached to the side frame **621**.

[0259] The retainer member **670** has the rear wall portion **671** disposed on the seat rear side of the side frame **621**, the folded-back wall portion **672** formed by folding back the right end portion of the rear wall portion **671** and sandwiching the rear end portion (rear flange portion **621***c*) of the side frame **621**, the side wall portion **673** bent from the left end portion of the rear wall portion **671** and extending to the seat front along sides of the airbag module **641** and the side frame **621**, and a front wall portion **676** bent from the front end portion of the side wall portion **673** and protruding to the inside in the seat width direction.

[0260] The retainer member **670** is attached to the side frame **621** from the outside in the seat width direction and is attached so as to sandwich a part (rear flange portion **621***c*) of the side frame **621**.

[0261] The retainer member **670** is disposed between the side frame **621** and the airbag module **641** in the seat width direction.

[0262] Specifically, the retainer member **670** (side wall portion **673**) is fastened together with the side frame **621** by the assembly shaft **644***a* provided on the inflator **644** and protruding from the inflator **644** toward the inside in the seat width direction.

[0263] The retainer member **670** further has the reinforcement portion **674** attached to the connection part between the side wall portion **673** and the front wall portion **676** and protruding from the side wall portion **673** to the inside in the seat width direction.

[0264] The reinforcement portion **674** is a reinforcement rib (screen portion) having a substantially triangular section and disposed at a position corresponding to the airbag module **641**.

[0265] Specifically, the reinforcement portion **674** is disposed at a position overlapping the airbag module **641** in the seat front to back direction and the up to down direction (the same position) and protrudes in a direction away from the airbag module **641**.

[0266] The reinforcement portion **674** is disposed at a position close to the inflation deployment starting point positions of the airbags **642** and **643**.

[0267] By providing the retainer member **670** as described above, it is possible to suitably receive pressure entailed by the inflation deployment of the airbags **642** and **643**.

Eighth Embodiment of Conveyance Seat

[0268] Next, a conveyance seat S**8** of an eighth embodiment will be described with reference to FIG. **22**.

[0269] It should be noted that description of content overlapping with the conveyance seats S**1** to S**7** described above will be omitted.

[0270] The conveyance seat S**8** includes a seat main body having a seat back **702**, side support members **730**, a side airbag device **740**, and a guide member **780** disposed between the side support member **730** and a side frame **721** in the seat front to back direction and guiding the inflation deployment directions of airbags **742** and **743**.

[0271] The side support members **730** are L-shaped linear members (specifically, wire members) and are attached by welding to the seat front parts of the right and left side frames **21**.

[0272] The side airbag device **740** includes an airbag module **741** having the first airbag **742**, the second airbag **743**, and an inflator **744**.

[0273] It should be noted that the side airbag device **740** may have only the first airbag **742** without having the second airbag **743**.

[0274] The guide member **780** is a plate-shaped member elongated in the up to down direction, is attached to the side support member **730**, and is disposed so as to cover the gap between the side support member **730** and the side frame **721** in the seat front to back direction.

[0275] Specifically, the guide member **780** is hooked on the side support member **730** by a plurality of attachment members (for example, attachment clips), which are not illustrated.

[0276] It should be noted that the guide member **780** may be attached to, for example, the side frame **721**.

[0277] By providing the guide member **780** as described above, it is possible to suitably guide the inflation deployment direction of the side portion of the seat back **702** when the airbag is inflation-deployed. As a result, the airbag can be deployed with speed.

[0278] In addition, the guide member **780** can be easily attached using the side support member **730**.

[0279] Although a vehicle seat used in an automobile has been described as a specific example in the above embodiments, the present invention is not particularly limited and can also be used for various seats such as two-wheeled seats for two-wheeled vehicles, seats for vehicles such as trains and buses, and seats for conveyances such as airplanes and ships.

[0280] In the above embodiments, the conveyance seat according to the present invention has been mainly described.

[0281] However, the above embodiments are merely examples for facilitating understanding of the present invention and do not limit the present invention. The present invention can be modified and improved without departing from the spirit thereof, and it is a matter of course that the present invention includes equivalents thereof.

REFERENCE SIGNS LIST

[0282] S**1** to S**8**: conveyance seat
[0283] **1**: seat cushion
[0284] **1**A: middle portion
[0285] **1**B: side portion
[0286] **1***a:* pad material
[0287] **1***b:* skin material
[0288] **2**, **102**, **202**, **302**, **402**, **502**, **602**, **702**: seat back
[0289] **2**A: middle portion
[0290] **2**B, **102**B, **202**B, **602**B: side portion
[0291] **2***a,* **102***a,* **202***a,* **302***a,* **602***a:* pad material
[0292] **2***aa,* **602***aa,* **602***ad:* pad through hole
[0293] **2***ab:* inner pad
[0294] **2***ac:* outer pad

[0295] **2***b*, **102***b*, **302***b*, **402***b*, **602***b*: skin material
[0296] **2***ba*, **302***ba*, **402***ba*, **602***ba*: skin burst-open portion
[0297] **2***bb*, **302***bb*, **602***bb*: inner skin material
[0298] **2***bc*, **302***bc*, **602***bc*: outer skin material
[0299] **602***c*: pad high-hardness portion
[0300] **3**: headrest
[0301] **3***a*: pad material
[0302] **3***b*: skin material
[0303] **3***c*: pillar
[0304] **4**: reclining device
[0305] **5**: height link device
[0306] **6**: rail device
[0307] **10**: cushion frame
[0308] **11**: cushion side frame
[0309] **12**: pan frame
[0310] **13**: rear connection frame
[0311] **14**: support member (elastic spring)
[0312] **20**, **120**, **220**, **320**, **420**, **520**, **620**, **720**: back frame
[0313] **21**, **121**, **221**, **321**, **421**, **521**, **621**, **721**: side frame
[0314] **21***a*, **621***a*: frame main body portion
[0315] **21***b*, **621***b*: front flange portion
[0316] **21***c*, **621***c*: rear flange portion
[0317] **22**, **522**: upper frame
[0318] **23**: lower frame
[0319] **24**: wire member (elastic wire)
[0320] **25**: support plate
[0321] **26**: pillar attachment member
[0322] **27**: protective member (protective plate)
[0323] **27***a*: plate main body portion
[0324] **27***b*: plate front portion
[0325] **27***c*: plate rear portion
[0326] **28**: second protective member (protective cap)
[0327] **30**, **130**, **230**, **330**, **430**, **530**, **730**: side support member
[0328] **31**: wire main body portion
[0329] **32**: wire extending portion
[0330] **40**, **140**, **240**, **340**, **440**, **540**, **640**, **740**: side airbag device
[0331] **41**, **141**, **241**, **341**, **441**, **541**, **641**, **741**: airbag module
[0332] **42**, **142**, **242**, **342**, **442**, **542**, **642**, **742**: first airbag
[0333] **43**, **143**, **243**, **343**, **443**, **543**, **643**, **743**: second airbag
[0334] **44**, **144**, **244**, **344**, **444**, **544**, **644**, **744**: inflator
[0335] **44***a*, **144***a*, **244***a*, **644***a*: assembly shaft
[0336] **44***b*, **244***b*: assembly nut
[0337] **244***c*: attachment groove
[0338] **145**, **245**, **645**: guide member
[0339] **145***a*, **245***a*, **645***a*: one end portion
[0340] **145***b*, **245***b*, **645***b*: the other end portion
[0341] **645***c*: attachment member
[0342] **146**, **246**: attachment bracket, attachment member
[0343] **146***a*: slit
[0344] **447**: holder member
[0345] **447***a*: one end portion
[0346] **447***b*: the other end portion
[0347] **150**, **250**: movable body
[0348] **151**: attachment portion
[0349] **152**: rotating member
[0350] **152***a*: rotating main body portion
[0351] **152***b*: protruding portion (protruding rib)
[0352] **360**, **460**, **560**: retainer member
[0353] **361**, **461**, **561**: rear wall portion
[0354] **362**: inside wall portion
[0355] **363**, **463**, **563**: outside wall portion
[0356] **564**: frame attachment portion
[0357] **670**, **770**: retainer member
[0358] **671**: rear wall portion
[0359] **672**: folded-back wall portion
[0360] **673**: side wall portion
[0361] **674**: first reinforcement portion, reinforcement portion
[0362] **675**: second reinforcement portion
[0363] **676**: front wall portion
[0364] **780**: guide member
[0365] G: gap

1. A conveyance seat, comprising:
a seat back serving as a backrest portion; and
a side airbag device attached to a side part of the seat back in a seat width direction in order to mitigate an impact applied from a side of a conveyance, wherein
the seat back includes
a back frame having side frames disposed on right and left sides in the seat width direction, and
a side support member attached to a seat front part of the side frame and protruding to a seat front side beyond the side frame in order to cause a side portion of the seat back to protrude to the seat front side,
the side support member elongatedly extends in an up to down direction along the side frame and is disposed to form a gap with the side frame in a seat front to back direction,
the side airbag device includes an airbag module attached to one side surface that is either an outside surface or an inside surface of the side frame,
the airbag module has
a first airbag inflation-deployed on the one side surface side of the side frame,
a second airbag inflation-deployed on the other side surface side of the side frame, and
an inflator supplying gas into the first airbag and the second airbag, and
the second airbag passes through the gap formed between the side frame and the side support member in the seat front to back direction and is inflation-deployed on the other side surface side of the side frame.

* * * * *